Chapter 1

Introduction

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Research Techniques

1. Research
   - Research Aim and Objectives
   - Features

2. Types of Research
   - Descriptive vs. Analytical Research
   - Applied vs. Fundamental Research
   - Quantitative vs. Quantitative Research
   - Conceptual vs. Empirical

3. 6Ps of Research
   - Purpose of Research
   - Product of Research

Research

- A search for knowledge
- A scientific and systematic search for significant information on a specific topic
- The purpose of research is to answer questions and acquire new knowledge
  - seeks the answer of certain questions which have not been answered so far and the answers depend upon human efforts
- For example: few years ago: did not know exactly the moon's south side?
  - make some assumptions about it
  - by efforts, now able to give concrete answer of the problem

Research

- Research answers only those questions of which the answers are not available in literature i.e., in human knowledge
  - an original contribution to the existing stock of knowledge
- most important process for advancing knowledge to promote progress

MOTIVATION IN RESEARCH

- What makes people undertake research?
  1. Desire to face the challenge of solving the unsolved problems
     - i.e., concern over practical problems initiates research
  2. Desire to get intellectual joy of doing some creative work
  3. Desire to be of service to society

Research Aim

- to improve research procedures through the refinement and extension of knowledge
  - to discover answers to questions through the application of scientific procedures
- each research study has its own specific purpose

Example:

"To investigate the effectiveness of machine learning algorithms in detecting cybersecurity threats in network traffic."

Research Objectives

1. To gain familiarity with a phenomenon or to achieve new insights into it
   - exploratory or formulative studies
2. To portray accurately the characteristics of a particular individual, situation or a group
   - descriptive studies
3. To determine the frequency with which something occurs or with which it is associated with something else
   - diagnostic studies
4. To test a hypothesis of a causal relationship between variables
   - hypothesis-testing studies

Exploratory or Formulative Study

- to gain familiarity with a phenomenon or to achieve new insights into it
- typically conducted when there is limited existing knowledge or understanding of the topic
- lay the groundwork for further research and may help in the formulation of hypotheses or the development of theoretical frameworks

Example

- To review existing literature on machine learning techniques applied to cybersecurity and network traffic analysis

Descriptive Study

- to portray accurately the characteristics of a particular individual, situation, or group
- focus on describing what currently exists or what has occurred, without attempting to determine causality or establish relationships between variables
- to provide a snapshot of a given phenomenon, helping researchers to understand its features, patterns

Example

- To identify common types of cybersecurity threats encountered in network traffic, such as malware, intrusion attempts, and data breaches
- To preprocess the dataset by extracting relevant features and preparing it for training and testing the machine learning models

Diagnostic Study

- To determine the frequency with which something occurs or with which it is associated with something else
- To identify and analyze relationships between variables, uncover patterns and diagnose problems or conditions

Example

- Conducting statistical analyses to determine the correlation between the performance of machine learning algorithms in detecting cybersecurity threats and characteristics of the network traffic data, such as traffic volume, data distribution, and network topology
- Identifying and defining key performance indicators (KPIs) for evaluating the effectiveness of machine learning algorithms in detecting cybersecurity threats, such as detection rate, false positive rate, and response time

Hypothesis testing Study

- To test a hypothesis of a causal relationship between variables
- To determine whether there is a statistically significant relationship between variables
  - ANOVA test
  - t-test

Case Study: Research Objectives

Software engineer wants to develop a e-commerce website. Design a step-by-step to fulfill research objectives

- **Exploratory or Formulative:** Conduct a preliminary investigation into the e-commerce industry, trends, and consumer behavior
  - Explore existing e-commerce websites to gain insights into design trends, features, and user experience
  - Research the technologies and platforms commonly used for developing e-commerce websites
- **Descriptive:** Define the target audience for the e-commerce website, including demographics, preferences, and buying behavior
  - Conduct market research to identify competitors, their products, pricing strategies, and unique selling propositions
  - Document the key features and functionalities required for the e-commerce website based on user needs and market analysis

- **Diagnostic:** Analyze existing e-commerce platforms or websites to identify common issues, challenges, and pain points faced by users
- **Hypothesis-Testing :**Formulate hypotheses about the potential impact of specific variables (e.g., website design, product pricing, marketing strategies) on user engagement and conversion rates
  - Design experiments or A/B tests to systematically test the hypotheses by manipulating variables and measuring outcomes
  - Refine and optimize the e-commerce website based on the findings from hypothesis-testing research to improve its effectiveness in achieving business objectives

Features of Research

1. **Systematic Process**
2. **Objective Orientation**
3. **Empirical Basis**
4. **Logical Reasoning**
5. **Cumulative Nature**
6. **Generalizability**
7. **Ethical Considerations**
8. **Communication and Dissemination**
9. **Continuous Improvement**

Systematic Process

- follows a systematic and structured process that involves:
  - formulation of a problem
    - research questions or hypotheses or problem statement
  - data collection
  - analysis
  - interpretation of findings

Example

- Conducting a study to investigate the impact of mindfulness meditation on stress levels among college students
- The research follows a systematic process
  - defining research questions
  - designing a randomized controlled trial
  - collecting data through surveys and physiological measures
  - analyzing the data using statistical methods
  - interpreting the findings

Objective Orientation

- driven by specific objectives, aims, or questions that guide the inquiry and provide clarity about the desired outcomes

Example

- Examining the relationship between sleep duration and academic performance among students
- Driven by specific objectives to determine if there is a correlation between the two variables (sleep duration and academic performance)

Empirical Basis

- empirical evidence obtained through observation, experimentation, or data collection from the real world
- emphasizes the use of verifiable and replicable methods to support conclusions and findings

Example

- Studying the effectiveness of a new teaching method on student learning outcomes in mathematics
- The research collects quantitative data on student performance scores before and after implementing the teaching method, providing empirical evidence to support conclusions about its efficacy

Logical Reasoning

- involves logical reasoning and critical thinking to analyze data, draw conclusions, and make inferences

Example

- Investigating the impact of environmental factors on public health outcomes, such as air pollution levels on respiratory illnesses
- The research applies
  - logical reasoning to analyze data
  - consider alternative explanations
  - draw conclusions about the causal relationship between environmental exposures and health outcomes

Cumulative Nature

- builds upon existing knowledge and contributes to the collective body of knowledge within a field or discipline

Example

- Examines the effects of rising global temperatures on agricultural productivity across different regions
- Findings contribute to the cumulative body of knowledge on climate change impacts
  - inform strategies for mitigating its effects on food security

Generalizability

- aims to generate findings that are applicable beyond the specific context or sample studied
- establish generalizable principles, patterns, or trends that can be applied to broader populations or situations

Example

- Examining the relationship between sleep duration and academic performance among students
- Findings based on a diverse sample of participants
  - provides insights into generalizable principles to find the relationship between sleep duration and academic performance

Ethical Considerations

- Research adheres to ethical principles and guidelines to ensure
  - the protection of human subjects
  - the responsible conduct of research
  - the integrity of the research process

Example

- Conducting a study on sensitive topics such as mental health or substance abuse
- Researchers obtain informed consent from participants
  - ensures the confidentiality of data
  - provide appropriate support or referrals for individuals experiencing distress

Communication and Dissemination

- Research findings are communicated through scholarly publications, presentations, and other channels to share knowledge, facilitate peer review, and contribute to the advancement of the field

Example

- Publishing research findings on the efficacy of a new teaching method on student learning outcomes in mathematics
- findings reach relevant stakeholders
  - contribute to advancements in teaching area

Continuous Improvement:

- Research is an iterative process that involves ongoing reflection, refinement, and revision
- Researchers continuously seek to improve their methods, theories, and approaches based on feedback, critique, and new insights

Example

- Revising research methods based on feedback from peer reviewers to enhance the validity and reliability of study findings
- Researchers may refine study designs, adjust data collection procedures, or incorporate additional analyses to address limitations and improve the quality of their research

Types of Research

1. **Descriptive vs. Analytical**
2. **Applied vs. Fundamental**
3. **Quantitative vs. Qualitative**
4. **Conceptual vs. Empirical**

Descriptive Research

- To portray accurately the characteristics of a particular individual, situation, or group
- focuses on describing what currently exists or what has occurred, without attempting to determine causality or establish relationships between variables

Methodology

- observational methods, surveys, questionnaires, and other data collection techniques to gather information about the subject of study
- emphasizes the collection of quantitative data to describe and summarize the features, behaviours, or attributes of interest

Analytical Research

- researcher has to use facts or information already available, and analyze these to make a critical evaluation of the material
- interpret data to uncover patterns, relationships, or causal links between variables
- seeks to understand why certain phenomena occur, predict future outcomes, or test hypotheses about the relationships between variables

Methodology

- employs statistical analysis, experimentation, modeling, and other advanced techniques to examine the relationships between variables and draw conclusions based on empirical evidence
- involves comparing and contrasting data, identifying trends, and testing hypotheses to infer causal relationships

Applied Research

- To find a solution for an immediate or practical problem facing a society or an industrial/business organisation
- Researchers collaborate with industry or government partners to tackle specific problems
- The results of applied research often lead to the development of new products, technologies, or processes

Fundamental Research

- also known as basic research, aims to deepen our understanding of fundamental principles, theories, and concepts
- driven by curiosity and seeks to explore the underlying mechanisms without immediate practical applications
- Researchers explore mathematical models, proofs, and abstract concepts

Quantitative Research

- involves collecting and analyzing numerical data to describe, predict, or control variables of interest
- aims to test causal relationships between variables, make predictions, and generalize results to wider populations

Qualitative Research

- understand the underlying meanings, experiences, and social contexts related to a phenomenon relating to or involving quality
- seeks to explore subjective experiences, beliefs, and behaviors

Conceptual Research

- also known as theoretical research, focuses on developing theories, models, or frameworks
- aims to deepen our understanding of fundamental principles and concepts
    - contributes to the theoretical foundation of a field

Empirical Research

- involves collecting and analyzing data to answer specific research questions
  - data-based research
- come up with conclusions which are capable of being verified by observation or experiment

6Ps of Research

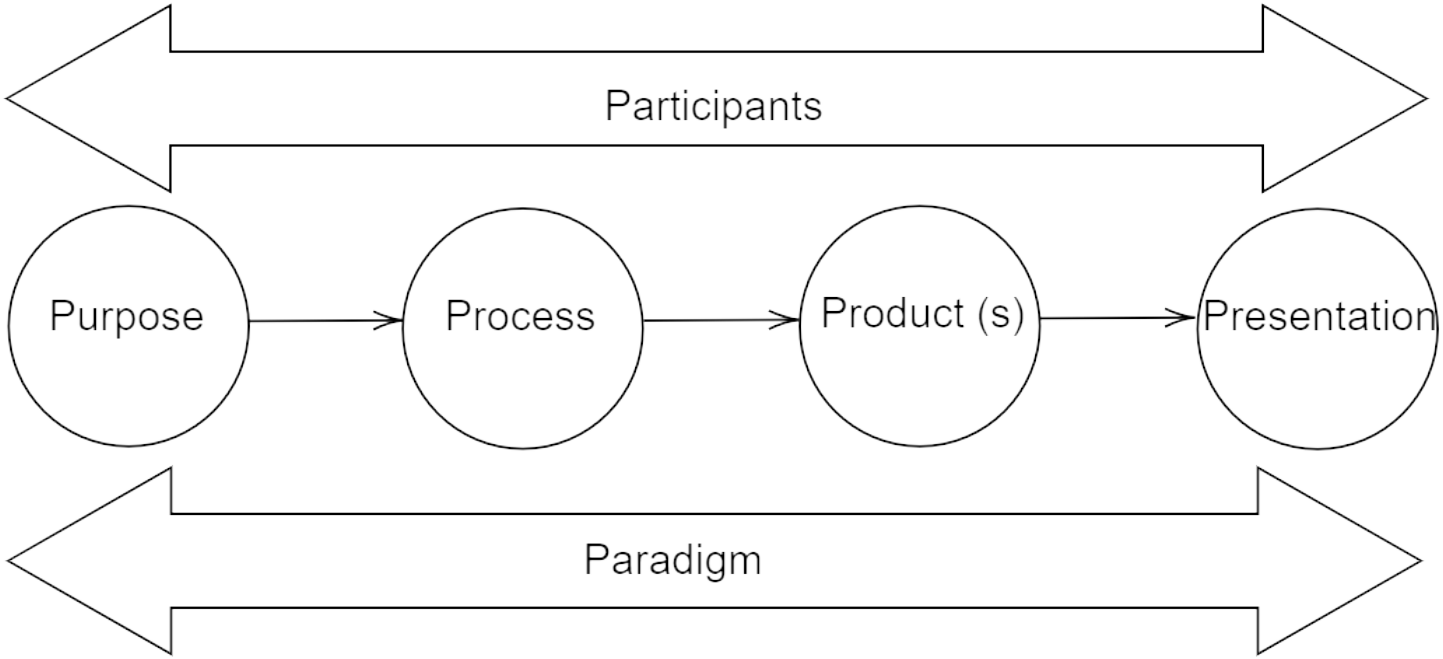
1. **Purpose**
2. **Product**
3. **Process**
4. **Participants**
5. **Paradigm**
6. **Presentation**

6Ps of Research

1. **Purpose:** the reason for conducting research
   - It clarifies what you aim to achieve and guides your research questions
   - why it is important or useful to study this, the specific research question(s) asked and the objectives set
   - Research without a purpose is unlikely to be good research
2. **Product:** outcomes of research, especially your contribution to knowledge about your subject area
   - include research papers, software prototypes, algorithms
   - Think about what you intend to produce as a result of your research effort
3. **Process:** the sequence of activities undertaken in any research
   - involves identifying research topics, establishing a conceptual framework (how you choose to think about your research topic)
   - selecting and using a research strategy and data generation methods, analyzing data
   - drawing conclusions, and recognizing any limitations in your research

4. **Participants:** individuals or groups involved in research
   - directly involvement: by interviewing them or observing them
   - indirectly involvement: editors to whom you submit a research paper
5. **Paradigm:** a pattern or model or shared way of thinking
   - that shapes research approach
   - need for a 'paradigm shift' to mean that a new way of thinking is required
6. **Presentation:** means by which the research is disseminated and explained to others
   - written up in a paper or thesis, or a conference paper is presented to an audience of conference delegates
   - a computer-based product is demonstrated to clients, users or examiners
   - Important that the presentation is carried out professionally
     - otherwise your audience might assume your whole research project was not undertaken in a professional manner

6Ps of Research



Purpose of Research

- incorporates the reason for doing it, the topic of interest
- why it is important or useful to study this
- the specific problem statement or research question(s) asked and the objectives set
- Why do people do research?
  - To add to the body of knowledge
  - To solve a problem
  - To find out what happens
  - To find the evidence to inform practice
  - To develop a greater understanding of people and their world

Product of Research

- A new or improved product
- A new theory
- A re-interpretation of an existing theory

Literature Review

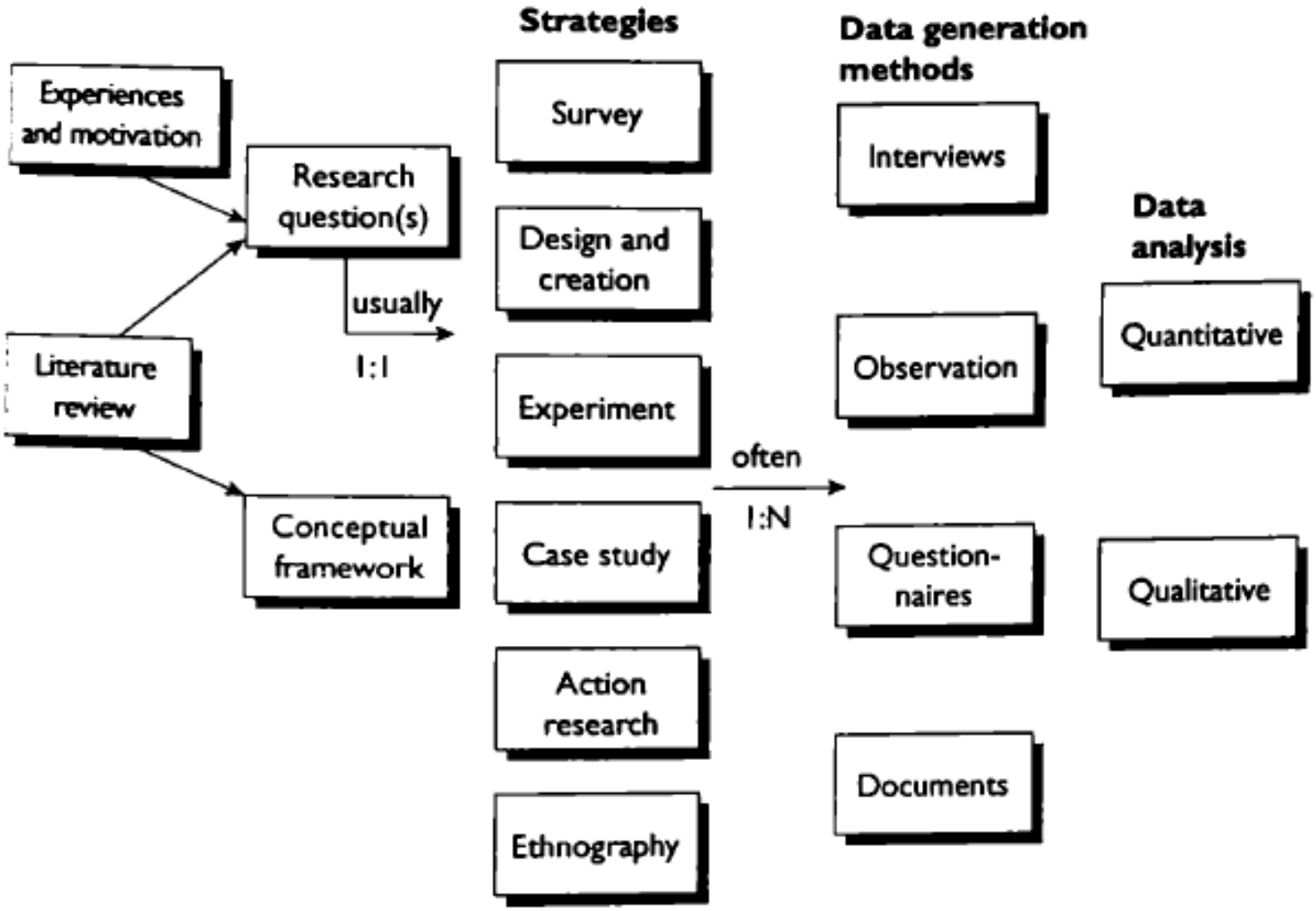
Chapter 2

Ashim Khadka

Research Techniques

1. Research Process
2. Literature Review
   - Literature Resources
   - Conducting a Literature Review
3. Research Question

Research Process



Literature Review

- **Reviewing related literature:** a crucial step in any research project
  - helps to understand the existing knowledge, identify gaps, and position your work within the scholarly context
- What has been done (searching, organizing, summarizing literature on a topic)?
- What needs to be done

Literature Review

- A literature review is an assessment of a body of research that addresses a research question
- An organized written presentation of what has been published on a topic by the scholars
- An Account of what has been published by accredited scholars and researchers

Objective of Literature Review

1. Demonstrate Awareness of Existing Work
   - shows that the researcher is familiar with prior studies related to their chosen topic
   - highlights the depth of understanding and awareness of the existing body of knowledge
2. Contextualize the Research
   - By placing the researcher's work within the context of what has already been published, the literature review establishes the relevance and significance of the new study
   - It helps readers understand how the current research contributes to the ongoing scholarly conversation
3. Evaluate Strengths, Weaknesses, and Bias
   - The literature review identifies strengths and weaknesses in previous work.
   - It also points out any biases or limitations present in the existing literature.

4. Identify Key Issues and Questions
   - Researchers use the literature review to pinpoint critical issues or unanswered questions in their field.
   - These unresolved matters guide the direction of the new research.
5. Address Research Gaps
   - The review identifies gaps—areas that have not been thoroughly explored or addressed by previous researchers
   - These gaps become opportunities for the current study to contribute new insights
6. Theories to Test or Explore
   - Researchers identify theories they will test or explore by collecting data from the field
   - The literature review informs the theoretical framework for the study
   - When researchers gather data, the literature review suggests theories that might explain their findings
7. Application Development
   - For computer applications, the literature review identifies relevant theories, genres, methods, or algorithms.
   - This informs the design and development process

8. Research Methods and Strategies
   - The review outlines the research methods or strategies the researcher will employ
   - It ensures transparency and rigor in the research process
9. Facilitate Understanding for Future Researchers
   - A well-structured literature review enables subsequent researchers to grasp the field's context and the significance of the current study

General Guidelines to Writing a Literature Review

- Don't attempt to cover everything written on the topic
- You need to pick out the research most relevant to the topic you are studying
- Only review work that is related to your topic and research questions or problem statement
- For new topics, review literature that is related to your topic

Roles of the Literature review

- A literature review helps establish the boundaries of the research area
  - By examining existing studies, you can identify what has already been explored and where gaps exist
    - gaps: help you to understand where you need to focus your efforts
  - Searching the literature involves reading and refining your problem
  - Understand what other scholars have asked, studied, and discovered
- Based on research gaps: formulate research questions or problem statement
  - determines which literature makes a significant contribution to the understanding of the research area or topic
- By reviewing the literature, you gain insights into research methodologies and tools that have proven effective
- Avoiding unintentional replication of previous studies

Steps for Writing your Literature Review

1. **Identify a research area or topic for investigation:** Begin by selecting a specific research topic or area
   - Consider what interests you and aligns with your field of study or research gap
2. **Locate the relevant literature:** Search for scholarly articles, books, and other academic sources related to your chosen topic
3. **Critically evaluate existing literature:** Read and analyze the literature you've gathered
   - Assess the quality, methodology, and relevance of each source
   - Look for gaps, contradictions, and areas where further research is needed
4. **Organize quality and relevant information:** Create an outline or framework to group similar studies together and identify key themes or trends
5. **Present the information effectively:** summarize the existing research
   - Highlight key findings, methodologies, and theoretical frameworks

Example: AI-based Intrusion Detection Systems

A literature review would help:

- Identify existing AI models used for cybersecurity.
- Understand challenges in real-time anomaly detection.
- Discover gaps, such as the lack of research on adversarial attacks in IDS.

Organization of the Review

1. **Introduction:**
   - Content - what is covered: focus on the main area
   - Structure - how it is organized: follow a chronological order, thematic grouping
   - Boundaries - what is outside of its scope
2. **Body:**
   - discussed and evaluated
     - Discuss the studies, theories, and concepts related to your topic
     - Evaluate the strengths, weaknesses, and relevance of each source
   - summarized and related to your research project
     - Summarize key findings from each study
     - Relate these findings to your own research project—how do they inform your work?
3. **Conclusion:**
   - Highlight the most relevant points: Recap the most significant findings or insights from the reviewed literature
   - Relate these back to the need for research
   - Reiterate what these mean for the research design

How To Write A Literature Review

- Formulate a problem
- Search
- Refine Topic
- Analyze, Construct Review
- Reference works

Literature Resources

1. Books
2. Journal articles
3. Conference papers and workshop proceedings

Books

- Textbooks can be useful as introductory sources: explain a field and the main approaches or theories used within it
  - gives guidance on particular methods or techniques
- Aimed at students on taught courses, not at academic researchers, which aspire
  - rarely cited in a literature review
- Some books, sometimes called '*monographs*' are aimed at an academic audience
  - can be cited in a literature review
  - theoretical rather than practical slant, do not contain exercises for the student to complete
  - authors can survey a field in depth develop ideas and discuss them in detail
- All books, whether textbooks or monographs: may be outdated even by the time of their first publication
  - contains theories or views that are out of line with current thinking

Academic journal articles

- Academic journal articles are where you should find information on the current thinking and research in your area of interest
- can be difficult to read at first, so you might find it is best to gain an understanding of the area through books before exploring journal articles for the latest developments
- Journal articles are
  1. Peer reviewed: academics unknown to an article's authors assess its suitability and quality before a decision is taken on whether to publish it in the journal
  2. Not peer-reviewed: can be biased towards the author's views or those of the organisation concerned.
- Some highly rated-journals
  1. IEEE Transactions
  2. ACM Transactions
  3. ScienceDirect
  4. MIS Quarterly

Conference and workshop proceedings

- Researchers will present their work initially via a paper at a conference or workshop before writing and submitting a more detailed journal article
- Conference and workshop proceedings: where you are likely to find the most up-to-date theories as well as suggestions for further research

Conducting a Literature Review

1. searching
2. obtaining
3. assessing
4. reading
5. critically evaluating
6. writing a critical review

Searching

- Scans the journals in your subject on an online database or a search engine
- To search resources you first have to define some keywords or search terms and then use them methodically to produce a list of potentially useful references
- Create a list of alternative terms for each concept, e.g., "selling" vs. "sales"
- Select an appropriate resource for your search such as book, online databases, catalogs, or search engines
  - Decide which source aligns with your research perspective
- Continue searching, using different combinations of search terms
  - Take advantage of features like symbols (e.g., AND, OR, NOT) for electronic searches

Obtaining

- Obtaining relevant sources is essential for building a solid foundation for your research
- Obtain full-text articles, books, or other materials through your institution's library or interlibrary loan services or online repositories such as arXiv, ResearchGate, or institutional repositories

Assessing

- Assessing the credibility of sources is indeed crucial when conducting a literature review
  - do not just take it at face value of literature review
- Ask questions yourself about books, journals, conferences
- Assess the **book** itself by asking:
  - Is the author someone eminent in the field? : well-regarded in the field
  - Have you heard of the publisher before? (MIT Press, Oxford University Press)
  - Is the book in a second or subsequent edition, or has it been reprinted?
    - Multiple editions or reprints indicate demand and relevance
    - A book that has been updated or reprinted suggests its enduring value

Assessing

- Assess the **journal article** itself by asking:
  - Is it aimed at academics or practitioners?
    - Focus on journals aimed at academics rather than practitioners
  - How long has the journal existed?
    - Established journals (with higher volume numbers) are generally respected
    - However, new journals can also be valuable.
  - Does the journal provide a list of its editorial board and advisors, and do these people seem to be of high standing?
    - If they are, then the journal is probably an acceptable resource
- Does the journal state its policy for reviewing articles? Are all articles peer-reviewed, some, or none at all?
  - You should concentrate on those articles that are peer-reviewed

Conferences and Workshops

For conferences and workshops, you need to ask:

- Is it an academic conference or one aimed solely at practitioners?
- Is the conference/workshop well established? As with journals, conferences held annually for many years are likely to be respected, but the converse is not necessarily true.
- Does the conference/workshop give details of its programme committee? Do they appear to be scholars of high standing?
- Does it make clear its reviewing policy, or were all papers accepted for presentation?

Reading

- Having obtained and assessed a literature source, you need to read it
- Reading and quickly extracting the necessary information from a text, often referred to as "**gutting**" a text, is a valuable skill, especially when working with a large volume of literature
- Structured approach is needed to effectively and efficiently gut a text:

1. For Journal Articles or Conference Papers:
   - Start with the abstract. It provides a concise summary of the entire article, including the research question, methods, results, and conclusions
   - Read the introduction to understand the context, background, and objectives of the research
   - Move to the conclusion to see the main findings and the significance of the study
   - Skim through the section headings to get a sense of the structure and main topics covered in the paper
   - Focus on the first and last sentences of each paragraph, as they typically contain the key points and summarizing ideas

2. For Books:
   - Review the table of contents to get an overview of the book's structure and main topics
   - Scan the index to identify specific topics of interest and to see how extensively they are covered
   - Read the preface or introduction to understand the purpose and scope of the book

Once you have gained an overview of a text's content, you can decide if it is necessary to read the whole thing

Critically evaluating

- Determine if the text provides information or insights that are directly applicable to your research.
  - Ask yourself why you are reading this paper and what you hope to gain from it
- Assess the quality and reliability of the data presented
- Are the methods used to collect and analyze the data appropriate and robust?
  - Are there parts you disagree with? Why?
- Evaluate whether the results are clearly presented and whether they convincingly support the conclusions
  - Is there some false logic or unwarranted assumptions?
- It can be difficult at first to have the confidence to criticize a paper or disagree with its content
- Identify any gaps in the research, such as areas
  - the author might have failed to address significant questions
  - Look for any methodological flaws or weaknesses that could impact the validity of the findings

Writing a critical review

- involves synthesizing and evaluating the existing literature on your research topic to demonstrate the relevance and necessity of your study
- This can be one chapter in a thesis, or one section in a paper, or it can be spread across your whole thesis/paper as you tackle different issues
- Link together the different ideas you have read in the literature into a coherent and persuasive argument for the value and necessity of your research
    - Organize the review around key concepts, themes, or issues rather than individual authors

Write the critical review in funnel approach (start board than narrow)

- Begin by discussing broad concepts and key ideas in the field
- Gradually narrow down to the specific research questions your study addresses
- Identify gaps in the existing literature and explain how your research will address these gaps
- Instead of summarizing each paper individually, synthesize the findings
  - Combine insights from multiple studies to build a comprehensive understanding of the topic
  - Cite multiple authors within the same theme to show the wide/breadth of research
- Evaluate the quality and relevance of the studies, pointing out strengths, weaknesses, and areas of controversy
- Create a matrix to map out which papers cover which concepts
  - This will help you organize your review thematically and ensure that you cover all relevant literature
- Follow the referencing style required by your institution, journal, or conference

Plagiarism

- Plagiarism is a serious academic offence that involves using someone else's words or ideas without proper attribution
- Plagiarism occurs when:
  - You use someone else's words or ideas without giving them credit
  - You present someone else's work as your own
  - You copy and paste text from electronic sources without proper citation
- How to avoid plagiarism
  - When you restate someone else's ideas in your own words, you still need to cite the original source
  - When you use an author's exact words, put them in quotation marks and cite the source
  - Familiarize yourself with the required citation style (e.g. IEEE, Harvard, APA, MLA, Chicago) and apply it consistently
  - By being careful to cite the paper:
    - you cannot be accused of plagiarizing
    - you demonstrate how aware you are of other published work

Citation Styles

- standardized ways of giving credit to the sources you have used in your writing
- Different academic fields and journals prefer different citation styles
- Three common citation styles:
  1. IEEE (Institute of Electrical and Electronics Engineers)
  2. APA (American Psychological Association)
  3. Harvard
- Always be consistent with the citation style you are using
- **In-text citations:** Provide brief details in the text and direct readers to the full reference list
- **Reference list:** Include full publication details at the end of your document, formatted according to the citation style guidelines
- **Check Guidelines:** Always check the specific guidelines of the journal or institution to ensure compliance with their preferred citation style

IEEE Citation Style

- citation style is commonly used in technical fields, particularly in electrical engineering and computer science
- uses numbered references in the text, corresponding to a numbered list of references at the end
- **In-text citation:** Numbers are placed in square brackets within the text such as [1]
  - Once citing several sources at once, it is recommended to list each number in separate brackets
  - A comma or a dash should be placed for separation. As an example: [2], [11], [34]. Alternatively, students can use [5] – [12]
  - Example:
    - "...end of the line for my research [13]."
    - "... as shown by Brown [4][5], as previously stated."
    - "The theory was first put forward in 1987 [1]."
    - "Several recent studies [3, 4, 15, 16] have suggested that..."

- **Reference list:** At the end of the document, the references are listed in the order they were cited in the text in a section headed References, in numeric order
  - Align references left
  - Place the number of entry at left margin, enclosed in square brackets
  - If there are six or more authors, then abbreviate them using '*et al*'
- **Reference examples**
  1. **Book**
     - [*Refno*.] Author's initials. Author's Surname, "Title of chapter in book," in *Book Title*(in italic), edition (if not first), Editor's initials. Editor's Surname, Ed. Place of publication: Publisher, Year, page numbers.
     - [2] C. W. Li and G. J. Wang, "MEMS manufacturing techniques for tissue scaffolding devices," in *Mems for Biomedical Applications,* S. Bhansali and A. Vasudev, Eds. Cambridge: Woodhead, 2012, pp. 192-217.

IEEE Citation Format

Bracketed numbers have to be placed before the start of each entry.

The author's name should be listed as First Initial and Last Name. For example, Einar Jonas Halldorsson should be cited as O. Jonas Halldorsson and not as Jonas Einar, Halldorsson.

[Number] First Name Initial(s) Last Name. "Page Title." Website Title. Web Address [accessed Month Day Year].

Articles have to be put in quotation marks.

Journal or book has to be in italics.

Example in Bibliography:

[8] D. Wilson. "Aeronautics Management and Psyche: Templates." Nevada Center Engineering Institute. http://www.site.org [accessed Feb. 14, 2018].

APA Citation Style

- Citation style is commonly used in the social sciences, such as psychology, sociology, and education
- In-text citation: The author's last name and the year of publication are included in parentheses
  - Basic Format: (Author's Last Name(s) or Organization, Year)
  - Example:
    - According to research, there are... (Wagner & Phelps, 2017)
    - visibility of acoustic waves becomes transparent and the waveforms are not affected (Wagner, 2012, p.104)
- Reference list: At the end of the document, references are listed alphabetically by the last name of the first author

APA 7th Citation Format

Surname, A., & Surname, B. (Year). Title. *Journal, Volume*. URL.

When you had three to five authors, it was necessary to list them all in APA 6th edition and only then use the "et al." part. Now the newest edition must use "et al." citing when you have three or more than three authors.

No need to include the publisher's location like a city or state after the publisher's name in a reference anymore. It had to be provided in APA 6.

When you had a secondary source citing, there was no need to include the date of your original source. The new addition requires it added.

The words "Retrieved from" or "Accessed from" are no longer necessary before the URL in APA 7.

Example in Bibliography:

Soranzo, A., & Grassi, M. (2014). Psychoacoustics: a comprehensive MATLAB toolbox for auditory testing. *Frontiers In Psychology, 5*. https://doi.org/10.3389/fpsyg.2014.00712

Harvard Citation Style

Harvard Citation Format

If you're interested only in one chapter of some book, you should cite it separately instead of referencing the entire source. Just add "Ch. 4" (or a number you need) at the very end.

Capitalize the document's title (unless it's an article).

Author, A. A., Year of publication. Title of Document. [medium] Name of Organization/Publisher. Available at: ‹link› [Accessed Day Month Year].

For many authors, list all authors in the order they're mentioned in the source itself. Put 'and' before the final name.

For journal articles, you're going to need volume and issue numbers. Remember that journals' and magazines' titles must be in italics while titles of articles shouldn't be formatted.

Example in Bibliography:

Loms Group Ltd., 2017. Use of Ethical Practices. [online] Loms Group Ltd. Available at: ‹http://www.loms.com/111› [Accessed 8 April 2019].

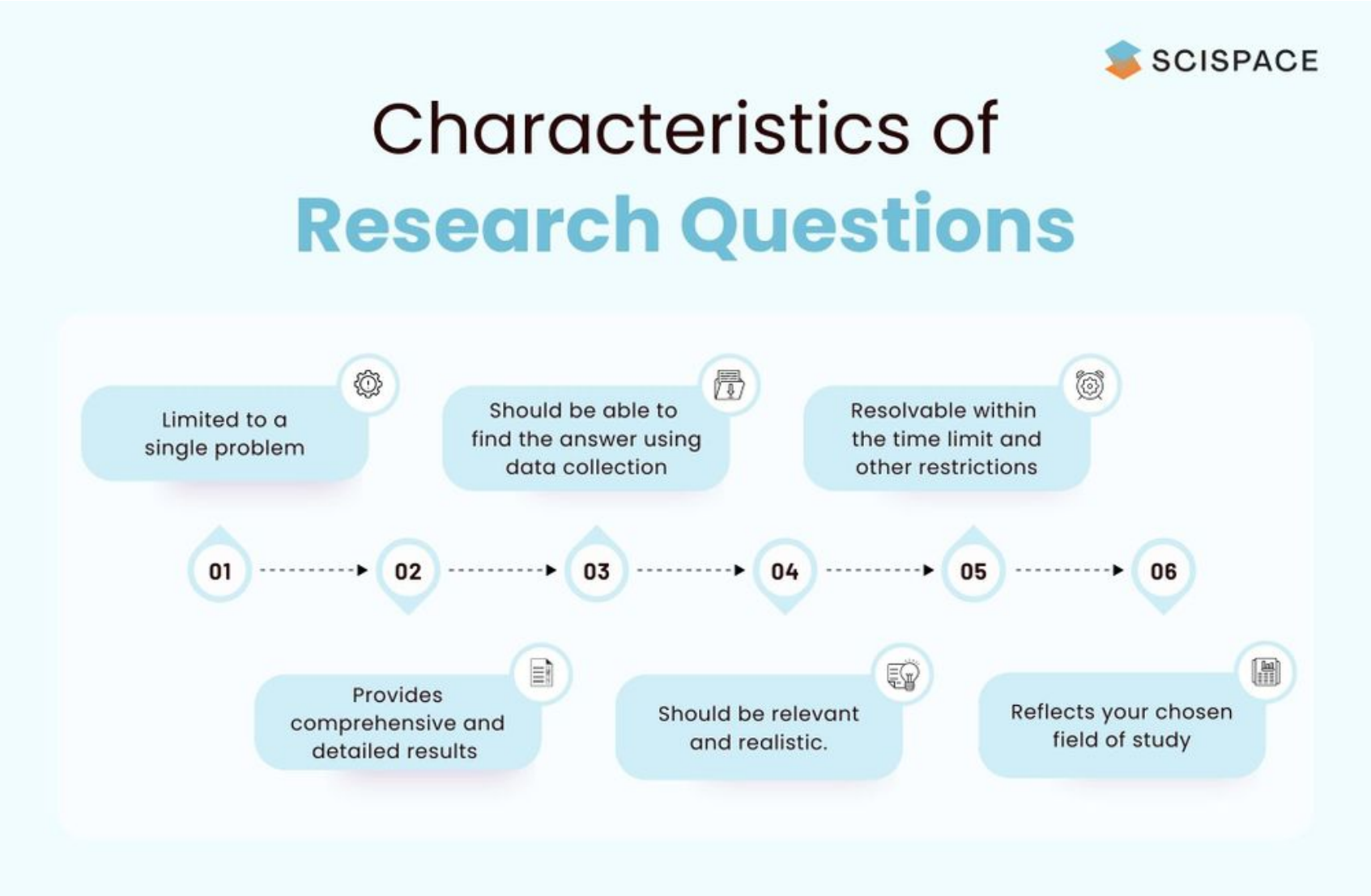
Example in Text:

According to Kostins (2009), it is crucial to research rare sea species.

Research Question

- A research question is an inquiry that the research attempts to answer
  - grounded on research
  - focus on a single research question
- It is the heart of the systematic investigation in any research project
  - narrows down a broad topic of interest into a specific area of study
- A research question is:
  - **Clear:** provides enough detail that the audience understands its purpose without any additional explanation
  - **Focused:** It is so specific that it can be addressed within the time constraints of the writing task
  - **Succinct/concise:** written in the shortest possible words
  - **Complex:** It is not possible to answer it with a "yes" or "no", but requires analysis and synthesis of ideas before somebody can create a solution
  - **Argumental:** Its potential answers are open for debate rather than accepted facts

The Path to a Research Question:
From Broad Topic to a Specific Question
Topic
Working knowledge of the topic
(narrowed down through preliminary research)
Working questions
(problematization, gap-spotting)
Research question/s
(after evaluation of questions)
Research.com


SCISPACE
Characteristics of
Research Questions
Limited to a single problem
Should be able to find the answer using data collection
Resolvable within the time limit and other restrictions
01
02
03
04
05
06
Provides comprehensive and detailed results
Should be relevant and realistic.
Reflects your chosen field of study

Characteristics of Research Questions

- Should include only one problem in the research question
- Should be able to find the answer using primary data and secondary data sources
- Should be possible to resolve within the given time and other constraints
- Detailed and in-depth results should be achievable
- Should be relevant and realistic
- It should relate to your chosen area of research

Example: Machine Learning for Fake News Detection

- Gap Identified: Existing studies focus on English news, but no research on fake news in regional languages.
- Research Contribution: Develop an AI model for detecting fake news in multiple languages.

Research Question

- A research question is a **specific inquiry** that the study aims to answer.
- It helps guide the research process by defining **what will be investigated**.
- Often Starts with "How," "What," or "Why" – Seeks explanation or exploration

Example: AI in Healthcare

Research Question:

- "How can deep learning models improve early diagnosis of lung cancer using CT scans?"

Problem Statement

- A problem statement describes the **issue or challenge** that the research aims to address.
- It explains **why** the research is important

Example: AI in Healthcare

Research Question:

- "Lung cancer is one of the leading causes of death worldwide, but early detection remains challenging. Traditional methods rely on radiologists, leading to delays and subjective diagnosis. There is a need for an AI-based automated detection system to enhance early diagnosis and improve survival rates."

problem statement explains the **issue** (late detection and subjective diagnosis) and **justifies the research** (need for AI solutions)

Example

Topic: "AI in Cybersecurity for Malware Detection"

Research Question:

"How can semi-supervised learning techniques improve malware detection accuracy while reducing the need for labeled training data?"

Chapter 3

Research Methodology

Ashim Khadka

Research Techniques

1. Research Methodology

2. Components of Research Methodology

3. Research Design
   - Experimental Research
   - Exploratory Research
   - Descriptive Research
   - Explanatory Research

4. Research Strategies
   - Experimental Research
   - Case Study
   - Action Research
   - Theoretical Research
   - Simulation-based Research

5. Multidisciplinary Research Approaches

Research Methodology

- the backbone of any scientific investigation, providing a structured approach to solving a problem or improving existing systems through investigation, experimentation, and analysis
  - in the field of computer science
- encompasses various stages, tools, and techniques to ensure that research is rigorous, repeatable, and valid

Components of Research Methodology

1. Research Problem
2. Objectives
3. Research Design
   a. Data Collection
   b. Sampling Techniques
   c. Data Analysis
   d. Hypothesis Formulation
4. Ethical Consideration
5. Validation and Reliability

1. **Research Problem**
   - the central question (primary work) or problem (secondary work) you aim to address
   - It's the "why" behind your study
     - A well-defined problem sets the direction and scope
     - It should be specific, measurable, and relevant to the field
   - Example:
     - In machine learning, identifying the most efficient algorithm for image classification
     - How can A algorithm reduce energy consumption in cloud computing systems?

2. **Objectives:** define what the research intends to achieve
Characteristics:
   - **Specific:** clear and unambiguous, outlining exactly what is to be accomplished
     - guides the research process by pinpointing what needs to be addressed, reducing ambiguity and focusing efforts
   - **Measurable:** measure progress and determine when the objective has been achieved
     - allows researchers to assess performance and make data-driven decisions, ensuring that progress can be quantified and evaluated
   - **Achievable:** be realistic, considering available resources, time, and technical capabilities
     - prevent overreaching and help maintain motivation and credibility
     - ensure that the goals are attainable given the research constraints
   - **Relevant:** must align with broader research goals and address a significant problem within the field
   - **Time-bound:** Setting a deadline or timeline for achieving the objective creates a sense of urgency and helps plan the research process
   - Example: Improving the accuracy of an AI-based facial recognition system.

3. **Research Design:** overall strategy or blueprint for conducting the research, including how data will be collected and analyzed—align with your objectives
   - design starts by revisiting the research problem and objectives, ensuring that the study's focus is clear
   - ensures that the chosen methods will directly address the questions or hypotheses at hand
   - **Methods:** Qualitative, Quantitative, or Mixed Methods
   - helps in planning resource allocation, including time, finances, and human resources, which are critical for conducting research effectively
4. **Data Collection:** specifies how relevant data will be gathered
   - through various techniques such as interviews, surveys, experiments, and observations

5. **Sampling Techniques:** methods to select a subset of data from a larger population.
   - outlines the techniques for selecting a representative subset of the population
   - whether using random, stratified, or purposive sampling, the design must justify the chosen approach to ensure the data is representative and unbiased
   - Example: Selecting 1000 tweets from a dataset of a million for sentiment analysis.
6. **Data Analysis:** a detailed plan for how the data will be analyzed once collected
   - Applying statistical and computational methods to derive insights for quantitative data
   - Thematic or content analysis for qualitative data
   - Programming languages that will be used to process and analyze the data
   - Example: Using regression analysis to predict customer behavior.

7. **Hypothesis Formulation:** Creating an assumption or prediction that is tested through research.
   - critical step in the research process that involves developing a clear, testable statement predicting the relationship between two or more variables
   - connects theoretical ideas with empirical observation by positing a relationship between variables
   - narrows down the scope of the research, directing attention to specific variables and their interrelations
   - informs the choice of research design, data collection methods, and analytical techniques
   - provides a basis for statistical tests that determine whether the data support or refute the prediction
   - Example: "An adaptive deep learning algorithm is expected to reduce energy consumption by at least 15%."

Hypothesis: Components of a Hypothesis

a. **Independent Variable (IV):** factor that is manipulated or varied in the study
   - Example: The type of algorithm used (adaptive deep learning algorithm vs. standard algorithm)

b. **Dependent Variable (DV):** outcome or effect that is measured in response to changes in the independent variable
   - Example: Mobile device energy consumption

- **Expected Relationship:** A clear statement on how the IV is predicted to affect the DV
  - Example: An adaptive deep learning algorithm is expected to reduce energy consumption by at least 15%.
- **Testability:** The hypothesis must be structured in a way that empirical observations or experiments can either support or disprove it.

Types of Hypotheses

a. **Null Hypothesis** (H_0)**:** States that there is no significant effect or relationship between the variables
   - Example: "There is no difference in energy consumption between mobile devices using an adaptive deep learning algorithm and those using a standard algorithm."

b. **Alternative Hypothesis** (H_1 **or** H_a)**:** Contradicts the null hypothesis by asserting that there is a significant effect or relationship
   - Example: "Implementing an adaptive deep learning algorithm will reduce mobile device energy consumption by at least 15% compared to standard algorithms."

- **Directional Hypothesis:** Specifies the direction of the expected effect (e.g., increase or decrease)
  - Example: "The adaptive algorithm will decrease energy consumption."
- **Non-directional Hypothesis:** States that there is a difference or relationship but does not predict the direction
  - Example: "There is a difference in energy consumption between devices using different algorithms."

Steps in Formulating a Hypothesis

a. **Identify the Research Problem:** Clearly define the issue you want to address.
   - Example: How can energy consumption be minimized in mobile devices using advanced algorithms?

b. **Review the Literature:** Investigate existing studies to understand current knowledge and gaps
   - Example: Studies may suggest that adaptive algorithms can optimize resource usage better than standard ones

c. **Define the Variables:** Specify which factors will be measured (dependent variables) and which will be manipulated (independent variables).
   - Example: IV - type of algorithm; DV - energy consumption.

d. **State the Expected Relationship:** Articulate how you believe the independent variable will affect the dependent variable.
   - Example: "The adaptive algorithm will reduce energy consumption by at least 15%."

e. **Ensure Testability:** Frame the hypothesis in measurable terms, ensuring that it can be empirically tested
   - Example: Quantify energy consumption reduction as a % improvement.

7. **Ethical Consideration:** Ensuring integrity, privacy, and transparency in research
   - Participants should be fully informed about the research purpose, procedures, potential risks, and benefits
   - must voluntarily agree to participate without any force
   - Researchers must protect personal and sensitive information
   - Data should be anonymized or encrypted to ensure that individuals cannot be identified from the results
   - Researchers are obligated to report findings truthfully, avoid fabrication or manipulation of data, and disclose any conflicts of interest
8. **Validation and Reliability:** Ensuring that research findings are consistent and reproducible
   - Validation involves confirming that the research methods and instruments are accurately measuring what they are intended to measure
   - Reliability refers to the consistency and repeatability of the research results

Research Design

- Research design is the framework that guides the research process
- determines how data will be collected, analyzed, and interpreted
- main types of research design are:
  1. Experimental Research
  2. Exploratory Research
  3. Descriptive Research
  4. Explanatory Research

Experimental Research

- involves manipulating one or more independent variables under controlled conditions to observe the effect on a dependent variable
- design aims to establish cause-and-effect relationships between variables
- Experiments are conducted in a setting where external variables can be minimized
- Subjects or units are randomly assigned to different groups (e.g., control and experimental groups) to ensure unbiased comparisons
- design tests specific hypotheses by comparing outcomes between groups
- Example: "Comparing the efficiency of different algorithms for sorting large datasets."

Exploratory Research

- used when the topic is new or not well-understood
- primary goal is to gather preliminary insights, generate hypotheses, and identify key variables
- design is flexible, often qualitative, and does not seek to provide conclusive answers but rather to explore possibilities
- aims to gain initial insights, identify potential research questions, and explore possible methods for further investigation
- Examples:
  - Exploring public perceptions of AI privacy concerns
  - Investigating the potential applications of quantum computing in cybersecurity

Descriptive Research

- aims to describe or define a phenomenon, providing a snapshot of the current state of affairs
- does not manipulate variables but rather observes and records data
  - Often involves surveys, case studies, or observations
  - Focuses on summarizing and describing data
- Example:
  - Describing patterns of internet usage during peak hours
  - Survey collects data on cloud usage patterns, types of services utilized (IaaS, PaaS, SaaS), and satisfaction levels

Explanatory Research

- seeks to understand the reasons behind a phenomenon
- goes beyond description to identify cause-and-effect relationships
- design often involves both quantitative and qualitative methods and employs statistical techniques to test relationships between variables
- Examples:
  - Imagine you are investigating why a specific deep learning model performs better on certain image datasets than others
  - Explaining why certain types of malware are more successful than others

Research Strategies

- approaches or methods that researchers adopt to conduct a study, gather data, and answer research questions or test hypotheses
- provide the framework for planning, executing, and evaluating research projects
- determine how a researcher will tackle the problem at hand and are chosen based on the nature of the research problem, the objectives, and the type of data needed
- selected strategy should align with the research goals and be feasible given available resources (time, funding, expertise)

Common Research Strategies

1. Experimental Research
2. Case Study
3. Action Research
4. Theoretical Research
5. Simulation-based Research

Experimental Research

- Systematic approaches designed to establish **cause-and-effect relationships** by manipulating variables in **controlled settings**
- Controlled experiments involve manipulating one or more independent variables while keeping all other factors constant
  - This isolation helps establish a direct cause-and-effect relationship
  - Example:
    - In computer engineering, you might test a new compression algorithm. You could vary the algorithm's parameters while keeping the input data constant to measure changes in compression speed and efficiency.
  - All possible combinations of the variables (factors) are tested, which allows for the evaluation of both main effects and interaction effects

- subjects (or experimental units) are randomly assigned to either a control group or one or more experimental groups
  - randomization minimizes selection bias and ensures that differences in outcomes are due to the experimental treatment
  - Example:
    - To compare two different machine learning models for spam detection, you randomly assign incoming emails to be processed by either Model A (control) or Model B (experimental). The performance metrics (like false positive rate) are then statistically compared to see if one model performs significantly better.

Case Study

- detailed examination of a specific subject in its real-world context
- case should be chosen based on its relevance to the research question and its potential to provide rich, detailed insights
- case might be an individual, a project, an organization, or a specific event
- Using various data sources (interviews, documents, observations) increases the reliability and depth of the findings
- **Example:** For the cybersecurity system case study, gather data by:
  - Conducting interviews with IT staff, security analysts, and system developers
  - Reviewing internal reports, system logs, and policy documents
  - Observing system operations in real time

- Analyze the qualitative data to identify recurring themes, patterns, and insights
  - involves coding the data and categorizing information based on key themes
- Compare the observed patterns against theoretical propositions or expectations to see if they align or diverge
- Example:
  - A case study on an AI-based cybersecurity system can reveal critical insights into the practical challenges of system integration, user adaptation, and performance measurement, which can guide future implementations in similar settings

Action Research

- collaborative and iterative process aimed at solving practical problems while generating knowledge
  - recognize a problem or opportunity for improvement in a real-world setting
  - "What are the challenges faced by users of our current software system?"
  - propose potential solutions or interventions
    - involves setting clear objectives, outlining steps for change, and deciding on data collection methods to evaluate the intervention's effectiveness
- involves active participation from stakeholders and researchers, focusing on real-world issues
  - gathered data is analyzed, and stakeholders reflect on the outcomes
  - step is crucial for understanding what worked, what didn't, and why
  - insights are used to refine the intervention until the desired improvement is achieved

Theoretical Research

- focuses on the abstract and mathematical foundations of computation
- involves developing and analyzing theories, models, and algorithms to understand the fundamental principles of computing
- conduct a thorough review of existing literature to identify gaps or unresolved problems in theoretical computer science
- familiarize yourself with the mathematical and computational foundations relevant to your research area, such as algorithms, complexity theory, or formal logic

- develop specific research questions or hypotheses that guide your investigation
  - "Can we prove that a certain problem is NP-complete?"
  - "How can we improve the efficiency of a particular algorithm?"
- establish a theoretical framework that underpins your research, drawing from relevant mathematical and computational theories
  - develop mathematical models or algorithms that address your research questions
  - validate your models or algorithms theoretically by demonstrating their consistency with established theories or by proving their correctness
  - compare your findings with existing research to highlight contributions and advancements

Simulation-based Research

- involves using computational models or simulated environments to study complex systems or phenomena
- particularly useful when real-world experimentation is impractical or impossible
- Formulate specific research questions that guide the simulation study
  - "How does network congestion affect the performance of a cloud computing system?"
- Develop a simulation model that accurately represents the real-world system or phenomenon
  - involve mathematical equations, algorithms, or software tools
- Execute the simulation and analyze the data to draw conclusions about the system's behavior, using mathematical methods to evaluate performance metrics or outcomes
- Validate the simulation model by comparing its outputs with real-world data or established theories

Multidisciplinary Research Approaches

- involves combining insights and methods from multiple disciplines to address complex problems
- approach is increasingly important in today's interconnected world, where challenges often require solutions that span traditional disciplinary boundaries
- by leveraging the strengths and insights of multiple disciplines, researchers can develop more holistic and innovative solutions
- Examples:
  1. **AI in Healthcare:**
     - Developing diagnostic tools that analyze medical images (like MRI scans) to detect diseases early
     - For instance, deep learning algorithms are trained on large datasets of medical images to assist doctors in diagnosing conditions such as cancer
     - This integration not only improves diagnostic accuracy but also speeds up the process, potentially saving lives by providing quicker intervention

Benefits of Multidisciplinary Research

- provides a more comprehensive understanding of complex phenomena by integrating diverse perspectives
- fosters the development of innovative solutions by combining different methodologies and knowledge bases
- particularly effective for addressing real-world problems that involve multiple facets, such as environmental issues, healthcare challenges, or cybersecurity threats

Multidisciplinary Research using Qualitative and Quantitative Methods

- combining qualitative and quantitative methods provide a comprehensive understanding of complex phenomena
- particularly useful in fields like health sciences, social sciences, and interdisciplinary studies
- **Qualitative Methods:** focuses on understanding meanings, experiences, and contexts through non-numerical data such as text, images, or observations
  - Common techniques include interviews, focus groups, case studies, and content analysis
  - Provides rich, contextual insights into phenomena, allowing for deeper understanding of social dynamics and individual experiences
- **Quantitative Methods:**involves numerical data and statistical analysis to identify patterns, trends, and correlations
  - Surveys, experiments, and statistical modeling are common techniques
  - Offers generalizability and precision, enabling researchers to make predictions and test hypotheses

- **Mixed Methods Approach:** Integrates both qualitative and quantitative methods to leverage their strengths and mitigate their weaknesses
  - Provides a more comprehensive understanding of research questions by combining detailed contextual insights with broad, generalizable findings
  - Enhances credibility by validating findings through multiple methods, ensuring that results are consistent across different data types

Chapter 4

Data Collection

Ashim Khadka

Research Techniques

1. Data Collection
   - Primary Data
   - Secondary Data
2. Data Collection Methods
   - Surveys
   - Experiment
3. Data Preparation
4. Data Sampling and Its Techniques
5. Evaluation and Validation of Data

Data Collection

- Data collection is the backbone of any research
- Data is the fundamental building block of information
  - can be numerical, textual, visual, or auditory
- involves gathering information from sources to address research questions/problem statements and test hypotheses
- Data refers to raw facts, figures, or observations that are unprocessed form, data doesn't carry meaning until it is organized and interpreted
- Types of Data
  1. **Quantitative Data:** Numerical values that can be measured or counted
     - e.g., temperature readings, survey scores
  2. **Qualitative Data:** Descriptive information that can be observed but not measured
     - e.g., interview responses, observations

Types of Data Collection

1. Primary Data
2. Secondary Data

Primary Data

- Data collected firsthand by the researcher for a specific purpose
- Methods: Interviews, surveys, experiments, direct observations
- Advantages: Tailored to the research problem; up-to-date and relevant
- Challenges: Often time-consuming and costly to obtain

Secondary Data

- Data originally collected by someone else for other purposes (e.g., government reports, published articles, historical records)
- Methods: Census data, academic journals, archival records, online repository
- Advantages: Readily available; cost-effective and sometimes covering long time spans
- Challenges: May not perfectly fit the current research objectives; issues with currency or accuracy.

Data Collection Methods

1. Surveys
2. Experiments
3. Questionnaires
4. Interviews
5. Observation
6. Experiments
7. Documents

Survey

- Surveys involve collecting data from a large group of people or events in a standardized and systematic manner
- Researchers use statistical analysis to identify patterns and generalize findings to a broader population than the target group
- Planning and Designing Surveys
  1. data requirements
  2. data generation method
  3. sampling frame
  4. sampling technique
  5. response rate and non-responses
  6. sample size

Data requirements

- data could be on topics directly associated with your research question(s), or it could be only indirectly related, such as demographic data (age, gender, education) about the respondents
- normally only gets one opportunity with the respondents in a questionnaire or interview-based survey
- Think ahead about how you'll analyze the collected data
  - Consider patterns, correlations, and interpretations

- **Data generation methods** commonly used in surveys are:
  - **Questionnaires:** design standardized sets of questions
    - usually self-administered
    - Design Considerations: Clear wording, logical order, mix of open and closed questions
    - Example: Employee satisfaction questionnaire distributed via email
  - **Interviews:** involve direct interaction between researchers and respondents
    - One-on-one or small group discussions to gather in-depth information
    - Types: Structured (fixed questions), semi-structured (flexible), unstructured (conversational)
    - Example: Interviewing participants about their shopping habits
  - **Observations:** collect data by directly observing behaviours, events, or processes without intervention
    - Example: Observing customer behavior in a store
  - **Document surveys:** examine existing documents (e.g., reports, manuals, code repositories) or might need to create them

Sampling Frame

- A sampling frame is a list or collection representing the entire population (participants, events, or relevant documents) from which you select a sample for your survey
  - If your research question is concerned with how an IT help desk handles queries from users
    - sampling frame might be the helpdesk's log of all requests

Experiment

- Try something out and find out what happens
  - taking a different route to work would reduce my journey time
- Experiment could be criticized
  - How can you be certain that there were no other factors affecting yesterday's journey that might have caused your shorter journey time?
- A strategy that investigates cause and effect relationships, seeking to prove or disprove a causal link between a factor and an observed outcome
- Researchers start by developing a theory about their topic of interest, which leads to a statement (called a hypothesis) based on the theory that can be tested empirically via an experiment
  - An experiment is then designed to prove or disprove the hypothesis

Characteristics of an Experiment-based Research

1. **Observation and measurement:** researchers make precise and detailed observations of outcomes and changes when a specific factor is introduced or removed
2. **Process of experimentation:** This involves:
   - Measurement or observation of a factor (initial conditions)
   - Manipulation of circumstances (introducing or removing the factor)
   - Re-measurement or re-observation of the factor to identify changes (outcomes)
3. **Proving or disproving relationships:** establish whether there is a causal relationship between factors
4. **Identification of causal factors:** Determining which factor (independent variable) causes changes in another (dependent variable), aiming to establish cause and effect

5. **Explanation and prediction:** Researchers use established theories or propose new ones to explain causal links between factors
    - If the experiment supports a hypothesis, the future outcomes can be predicted based on the identified causal relationship
6. **Repetition:** Experiments are repeated numerous times under varying conditions to ensure that observed outcomes are consistent and not influenced by external factors like faulty equipment or chance

Data Preparation

- Data preparation ensures data is clean and ready for analysis
- Data Cleaning: Handle missing values, remove outliers, correct errors
- Data Coding: Convert qualitative data (e.g., "yes/no") into numerical formats for analysis
- Data Organization: Format data for software compatibility, ensure consistency
- Poor preparation can lead to inaccurate results, so pilot testing is recommended

Data Sampling and Its Techniques

- how to select actual sample (i.e., people or events) from the sampling frame (i.e., population)
- Allows generalization to the larger population if done correctly
- Types of Sampling Techniques
  1. **Probability sampling** involves selecting a sample from a population in a way that each member of the population has an equal opportunity to be chosen as part of the sample
  2. **Non-probability sampling** does not involve random selection and does not guarantee that each member of the population has an equal chance of being included in the sample

Table: Sampling techniques

Probabilistic	Non-probabilistic
Random	Purposive
Systematic	Snowball
Stratified	Self-selection
Cluster	Convenience

Probability Sampling Techniques

- **Random sampling:** People or events are selected literally at random
  - For example, drawing names from a hat or using random numbers from a list
- **Systematic sampling:** this builds on random sampling by adding a system of choosing people at regular intervals in the sampling frame
  - For example, if using the telephone directory, you might decide to choose every 100th person
    - researchers would not start with the first person in the directory but would choose the first name at random and then carry on from there
- **Stratified sampling:** randomly select a sample from each subgroup (strata) proportional to its size in the population
  - Ensures all subgroups are represented in the sample
  - Improves the accuracy of estimates for each subgroup
  - Example: Surveying employee stress in an IT company with 70% males and 30% females
    - The sample would have 70% males and 30% females chosen randomly from respective lists

Cluster Sampling

- Sampling methods in which the population is divided into clusters that naturally occur (towns, neighbourhoods, schools)
- After identifying clusters, a subset of these clusters is randomly selected
- Once clusters are selected, all individuals within the chosen clusters are sampled using another method (simple random sampling)
- By clustering, researchers can reduce costs associated with travel and data collection
- There is a risk of bias if clusters are not representative of the population or if there's significant variability between clusters
    - Random selection of clusters helps mitigate this but doesn't eliminate it entirely

Non-probabilistic Sampling Techniques

- **Purposive sampling:** researcher deliberately selects samples (individuals or cases) that are likely to produce valuable data to meet the purpose of the research
  - Purposive sampling is a valuable technique in qualitative research where depth and specificity are prioritized over statistical representativeness
  - Since participants are selected based on researcher judgment, there is a risk of bias if the selection criteria are not carefully defined or applied consistently
  - In a survey of attitudes to online purchasing, you might deliberately choose people you know who regularly be online and people who have never shopped online

Snowball Sampling

- sampling method used in qualitative research where the researcher starts with one or a few known participants and then asks them to suggest others who fit the criteria for inclusion in the study
- After gathering data from these participants, the researcher asks for further names, so that the sample soon snowballs in size
- Allows researchers to reach populations that are difficult to identify or access through conventional sampling methods
- Snowball sampling can introduce bias because participants are not randomly selected from the population
  - Participants may refer others who are similar to themselves, potentially limiting diversity of perspectives
- In the context of studying computer hackers, a researcher might start by identifying and interviewing one hacker known
  - researcher could ask this participant to recommend others within their hacker community who might also be willing to participate

Self-selection Sampling:

- Sampling occurs when individuals voluntarily choose to participate in a study based on their own interests or motivations.
- Researchers typically advertise their study and invite potential participants to respond if they are interested
  - can lead to a sample that is biased towards individuals with specific perspectives or experiences
- This method is particularly useful when researchers have limited access to the target population or when the population is difficult to reach through other means

Convenience Sampling

- Sampling involves selecting participants who are readily available or easy to access
- Researchers choose individuals based on their convenience rather than through random selection or a systematic approach
- Convenience sampling is appropriate in preliminary studies, pilot research, or when exploring initial hypotheses
- This method is often used for its practicality and efficiency, particularly when time and resources are limited
- Example: A researcher conducting a survey on mobile phone usage might approach people at a shopping mall for their opinions, resulting in a sample that may not represent the entire population's mobile phone habits

Evaluation and Validation of Data

- After collecting and preparing your data, you must evaluate its quality and validate its accuracy
- Evaluation involves assessing the overall quality of the data, including its accuracy, completeness, consistency, and relevance to the intended analysis
- Validation is a more specific process that checks whether the data meets predefined criteria or standards, such as correct formats, ranges, or cross-verification with external sources

Data Quality Evaluation

- **Reliability:** Consistency of data collection (e.g., same results if repeated)
- **Validity:** Accuracy, ensuring data measures what it's supposed to
  - checking for errors such as typos, incorrect entries, or outliers
  - For instance, if a customer ages includes an entry of 150, error can occur
- **Completeness:** Identifies any missing key information, which is critical for analysis
  - if so, decide on imputation or exclusion

Validation Techniques

- **Cross-Validation:** Cross-check data with other independent sources
  - Example: Validate online survey results by comparing them with customer purchase records.
- **Pilot Testing:** Run a small-scale pilot study to ensure your data collection instruments work correctly
  - Example: Conduct a pilot interview among 5 participants to refine your interview guide
- **Triangulation:** Use multiple data collection methods to confirm findings
  - Example: Combine observations with survey data to validate consumer behavior trends

Chapter 5

Data Analysis

Ashim Khadka

Research Techniques

1. Data Analysis

2. Types of Data Analysis
   - Quantitative Data Analysis
     - Types of quantitative data
   - Data Coding
   - Data Presentation
   - Analysis Techniques
     - Descriptive Statistics
     - Inferential Statistics
     - Regression Analysis
     - Time Series Analysis
     - Multivariate Analysis

3. Qualitative Data Analysis
   - Qualitative Data
   - Analyzing Textual Data
   - Analyzing Non-Textual Data

Data Analysis

- process of systematically applying statistical and logical techniques to describe, organize, summarize, and interpret collected data
- helps researchers identify patterns, test hypotheses, and derive meaningful conclusions to support decision-making and theory-building
- In research, data analysis allows scholars to:
  - Extract meaningful insights from collected data.
  - Identify patterns and trends to support or refute research questions.
  - Ensure reliability and accuracy in their findings.
  - Make informed decisions based on statistical evidence.

Purpose of Data Analysis in Research

- Summarization helps convert raw data into meaningful insights by organizing it into structured formats, such as: Tables, Graphs, Charts, and Statistical summaries (mean, median, mode, standard deviation, etc.)
  - Example: A researcher collects student test scores from 10 schools to analyze learning outcomes. Instead of listing thousands of scores, they can summarize:
    - Mean score: The average performance of students.
    - Median score: The middle value to avoid extreme values affecting results.
    - Standard deviation: The variation in student performance
- researchers must **interpret** the results to derive meaningful conclusions
  - Identifying patterns, correlations, or anomalies.
  - Comparing results against the hypothesis.
  - Understanding the real-world significance of data.

- Validation ensures that the research findings are accurate, reliable, and free from errors or biases
  - Checking for missing data (e.g., incomplete survey responses)
  - Verifying data consistency (e.g., ensuring all ages are within a reasonable range)
  - Cross-checking data sources (e.g., validating online data with official reports)
- Hypothesis testing is used to determine if research findings are statistically significant
  - Formulating a null hypothesis (H_0) (assumption that there is no effect)
  - Testing against an alternative hypothesis (H_1) (assumption that there is an effect)
  - Using statistical tests (e.g., t-test, chi-square test, ANOVA)
- Data-driven decision-making is crucial in research
  - Draw conclusions that answer research questions
  - Provide recommendations for future research or policy changes
  - Improve existing processes based on evidence

Steps in Data Analysis

1. Data Collection - Gather data through experiments, surveys, or observations
2. Data Cleaning - Remove missing values, inconsistencies, or outliers
3. Data Processing - Convert data into a structured format
4. Exploratory Data Analysis (EDA) - Identify trends, distributions, and patterns
5. Statistical Analysis - Use statistical tests to validate hypotheses
6. Data Interpretation - Draw conclusions based on results
7. Reporting - Present findings through tables, graphs, and research papers

Types of Data Analysis

1. Quantitative Data Analysis
2. Qualitative Data Analysis

Quantitative Data Analysis

- involves numerical data and statistical methods to uncover patterns, relationships, and trends
- includes understanding data types, coding for analysis, presenting data visually, and applying statistical techniques
- data generated by experiments and surveys
- descriptive statistical techniques allow more patterns to be found such as the average value of a set
- Types of quantitative data:
  1. Discrete Data
  2. Continuous Data
  3. Nominal Data
  4. Ordinal Data
  5. Interval Data
  6. Ratio Data

Discrete Data

- Consists of distinct, separate values (e.g., number of students in a class: 20, 21, 22)
- Cannot take intermediate values
- Example: Number of cars in a parking lot
- Analysis: Suitable for frequency tables and bar charts

Continuous Data

- Can take any value within a range (e.g., height: 1.75m, 1.76m, etc.)
- Measured rather than counted
- Example: Temperature readings
- Analysis: Use histograms, mean, and standard deviation

Nominal Data

- Categorical data without order (e.g., gender: male, female)
- Used for labeling variables
- Example: Eye color categories
- Note: Cannot calculate mean; use mode or frequencies

Ordinal Data

- Categorical data with a natural order (e.g., satisfaction level: low, medium, high)
- Differences between categories are not necessarily equal
- Example: Rating scales (1-5 stars)
- Note: Can calculate median, but mean may be inappropriate; suitable for non-parametric tests like Mann-Whitney U test

Interval Data

- Numerical data with equal intervals but no true zero (e.g., temperature in Celsius)
- Example: IQ scores, calendar dates
- Note: Can calculate mean and standard deviation, but ratios are meaningless (e.g., 20°C is not twice as hot as 10°C)

Ratio Data

- Numerical data with equal intervals and a true zero (e.g., weight, time)
- Example: Height in meters, income in dollars
- Note: Allows for all statistical calculations, including ratios (e.g., 10 kg is twice as heavy as 5 kg)

Data Coding

- process of converting raw data into a format suitable for statistical analysis, often by assigning numerical values to categories
- Example: If a survey has a question on satisfaction level
  - Very Satisfied $\rightarrow$ 5
  - Satisfied $\rightarrow$ 4
  - Neutral $\rightarrow$ 3
  - Dissatisfied $\rightarrow$ 2
  - Very Dissatisfied $\rightarrow$ 1
- conversion allows statistical analysis using numerical methods
- Avoid duplicating codes or mislabeling categories
- Handling missing data: delete or data imputation

Steps involved in Data Coding

1. Identify Variables: Determine what data needs coding (e.g., responses to a survey with categorical options)
2. Assign Codes:
   - For nominal data: Male = 1, Female = 2.
   - For ordinal data: Low = 1, Medium = 2, High = 3.
3. Create a Codebook: Document the coding scheme for reference, ensuring consistency (e.g., "1 = Male, 2 = Female").

Data Presentation

- Once data is collected, it needs to be visualized effectively
- Visuals make data easier to interpret and communicate, aiding in identifying patterns or trends
- Types:
  - Tables
  - Bar charts
  - Scatter Graph
  - Line graph

Tables

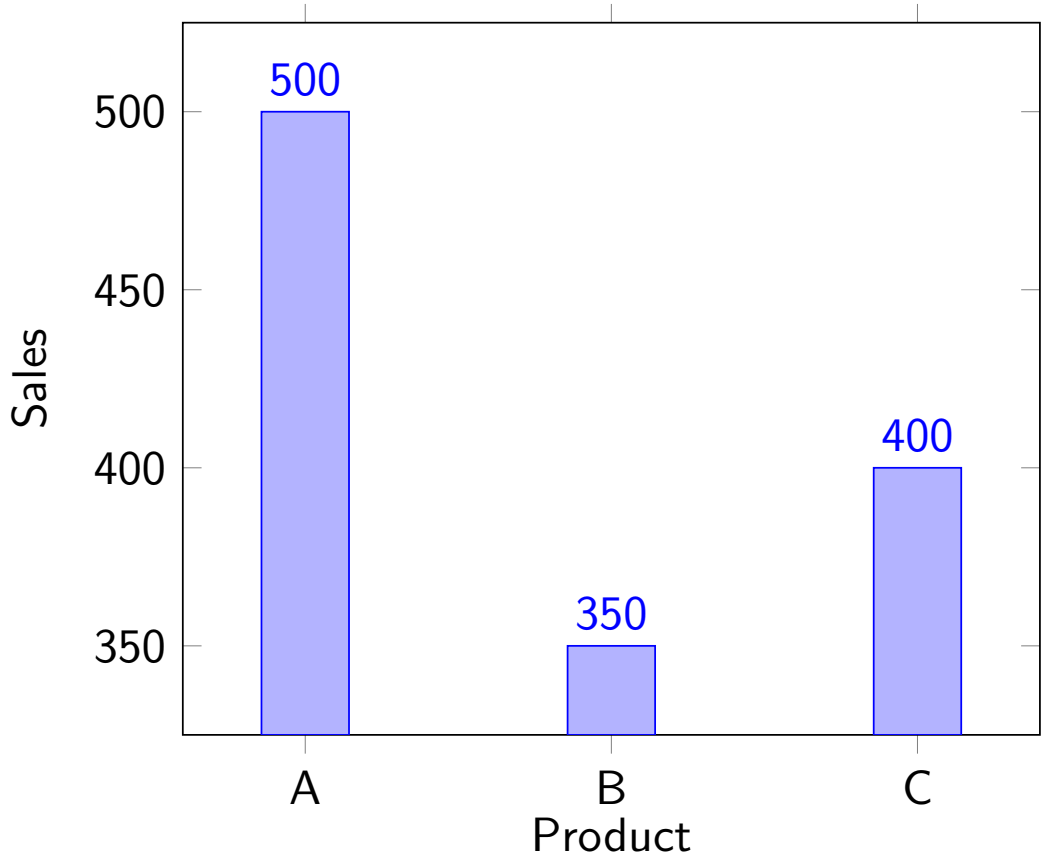
- Organize data in rows and columns for exact values
- Best for showing detailed data
- Use: Compare multiple variables or present raw data

Table: Sales of Different Products

Product	Sales
A	500
B	350
C	400

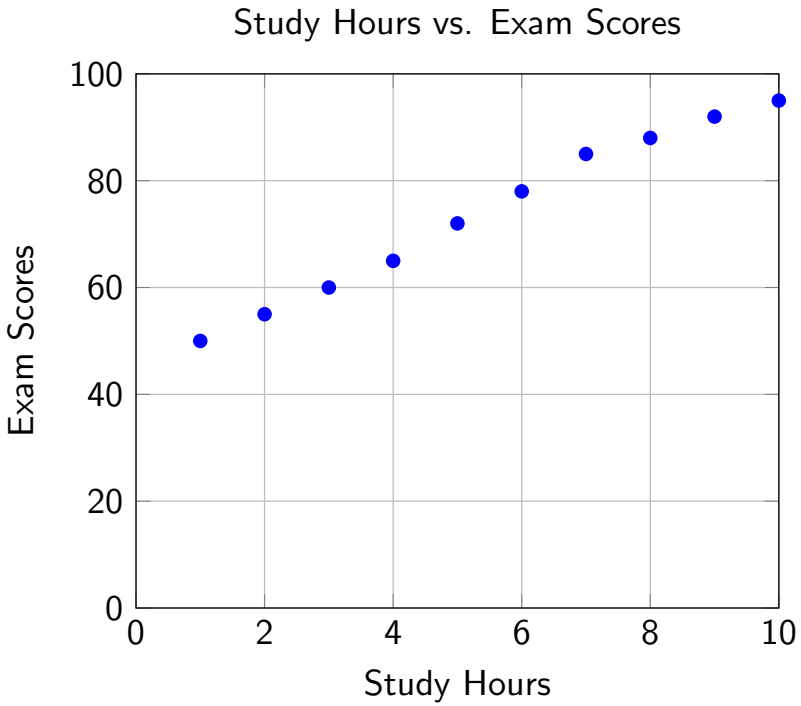
Bar Charts

- Use bars to represent categorical data
- Vertical or horizontal; gaps between bars indicate discrete categories
- Note: Preferred over pie charts for comparing categories due to easier visual comparison



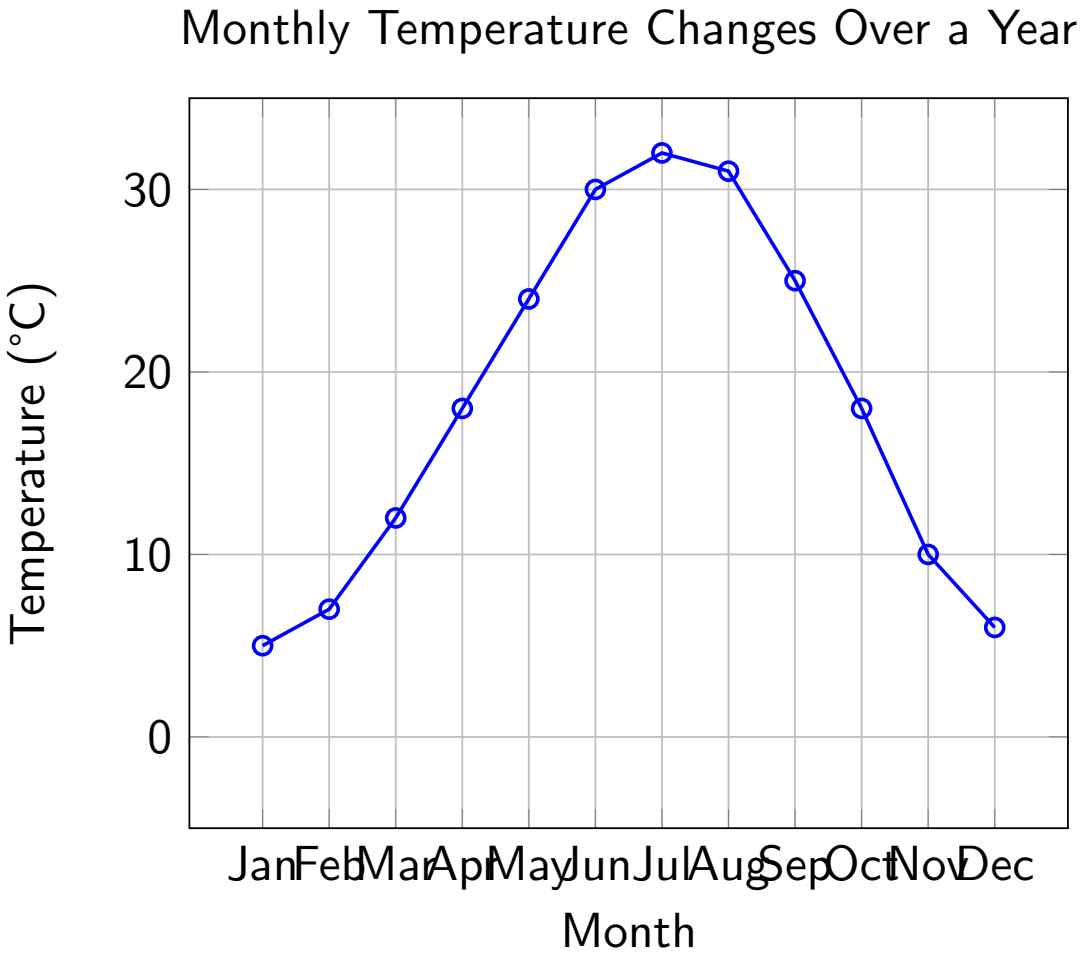
Scatter Graph

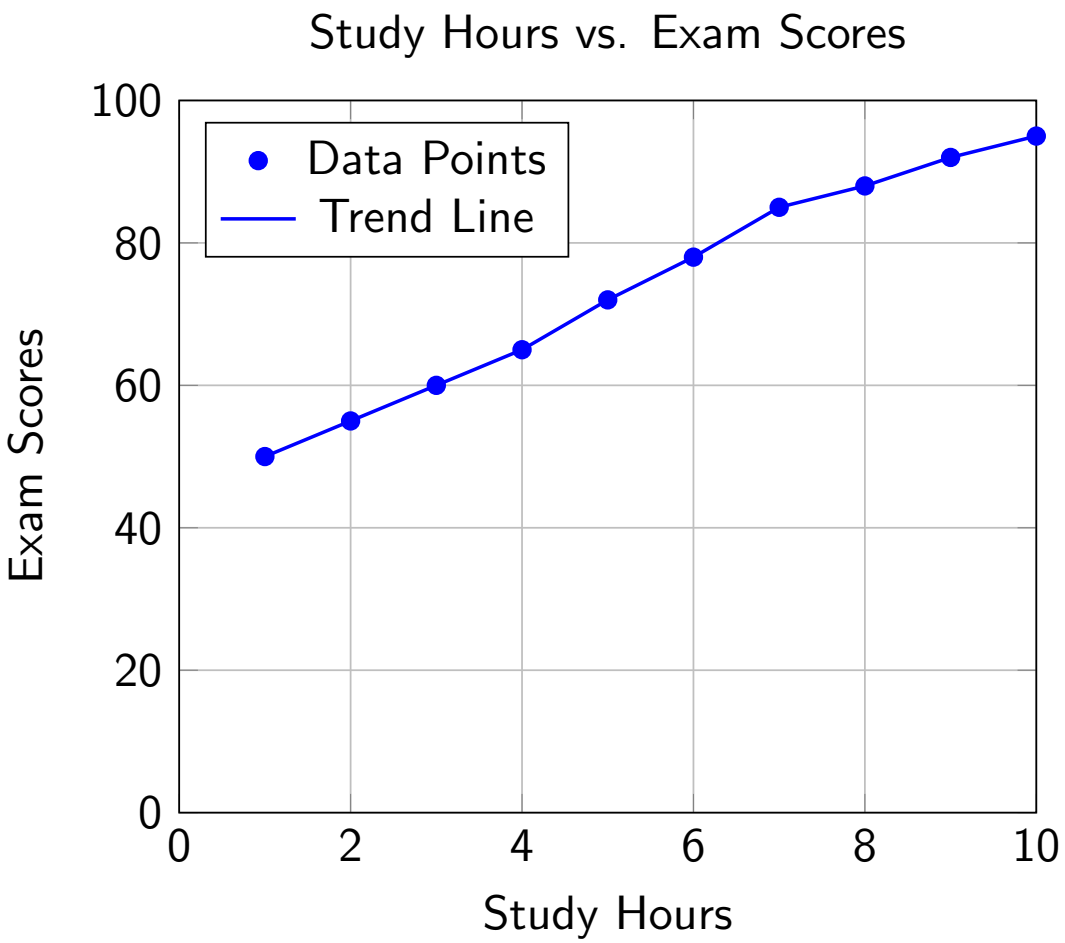
- Used to show relationships between two numerical variables
- Plot individual data points on X-Y axes to show relationships (e.g., height vs. weight)
- Useful for correlation analysis, identifying trends or outliers
- Example: Plot study hours vs. exam scores to see if more study correlates with higher scores



Line Graph

- Best for continuous data, especially time series
- Connect data points to show trends
- Example: Show monthly temperature changes over a year




Study Hours vs. Exam Scores
Data Points
Trend Line
Exam Scores
Study Hours
0
20
40
60
80
100
0
2
4
6
8
10

Analysis Techniques

1. Descriptive Statistics
2. Inferential Statistics
3. Regression Analysis
4. Time Series Analysis
5. Multivariate Analysis

Descriptive Statistics

- Summarize data to describe central tendency and variability
  - Use: Understand the dataset's basic characteristics without inference.
- Measures: Mean (average), median (middle value), mode (most frequent), standard deviation (spread), variance, range
  - Mean is sensitive to outliers; median is more robust for skewed data
- Example: Average test score $= 75$, standard deviation $= 10$, indicating scores vary by 10 points around the mean.

Inferential Statistics

- Draw conclusions about a population from a sample, testing hypotheses
- Used to make predictions or test hypotheses
  - Z-test: Compares means when the population variance is known
  - T-test: Compares means of two groups
  - Chi-square test: Analyzes categorical data
  - ANOVA: Compares means of multiple groups

Z-test

- Large samples, known population parameters/variance
- Compare sample mean to the population mean when the population variance is known
- Example: Test if a new drug improves recovery time, assuming known population recovery time variance

$$Z = \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}}$$

where $\bar{X}$: sample mean, μ : population mean, σ : population standard deviation, n : sample size

T-test

- Compare means of two groups (e.g., male vs. female test scores) when the population variance is unknown or the sample size is small ($n < 30$)
- relies on t-distribution
- helps to determine if differences between groups are statistically significant

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\bar{X}_1, \bar{X}_2$: sample means, s_1^2, s_2^2 Sample variances, n_1, n_2 : sample sizes

Types of T-Tests

1. One-Sample T-Test:
   - Compares the mean of a sample to a known population mean
   - Example: Testing if the average IQ of students in a class ($\bar{X}$ differs from the national average (μ)
2. Independent Two-Sample T-Test (Unpaired):
   - Compares the means of two independent groups
   - Example: Comparing male vs. female exam scores
3. Paired T-Test (Dependent Samples):
   - Compares means before and after an intervention on the same group
   - Example: Measuring student scores before and after using a new teaching method

ANOVA (Analysis of Variance)

- For Multiple Groups
  - More than two groups, to avoid multiple t-tests
- Compare means across multiple groups
- Examples: test scores by teaching method: lecture, group work, online

Chi-square test

- For Categorical Data
- Used to test relationships between two categorical variables
  - gender and voting preference
- Types: Test of independence (are variables related?), goodness of fit (does data fit expected distribution?)

Regression Analysis

- Model relationships between dependent and independent variables.
- Types:
  - Linear regression ($y = mx + b$, e.g., predict sales from advertising spend)
  - multiple regression ($y = b_0 + b_1x_1 + b_2x_2$, e.g., predict grades from study hours and attendance)
- Example: Predict student grades based on study hours, showing a positive slope (more study, higher grades)
- Use: Prediction, understanding variable relationships, assuming linearity

Time Series Analysis

- Analyze data collected over time (e.g., monthly sales trends)
- Components: Trend (long-term direction), seasonality (regular patterns, e.g., holiday sales spikes), noise (random variation)
- Example: Forecast next quarter's revenue using past sales, identifying seasonal peaks
- Use: Forecasting, trend analysis, requires time-ordered data

Multivariate Analysis

- Analyze multiple variables simultaneously to find patterns or relationships
- Techniques: Factor analysis (reduce variables to underlying factors), cluster analysis (group similar observations), discriminant analysis (classify into groups)
- Example: Group customers by income, age, and spending habits using cluster analysis for targeted marketing
- Use: Complex datasets, high dimensionality, to uncover hidden structures

Qualitative Data Analysis

- Qualitative data analysis focuses on non-numerical data, such as text or images, to uncover themes, meanings, and patterns
- nature of qualitative data and techniques for analyzing textual and non-textual forms

Qualitative Data

- Qualitative data is descriptive, non-numerical data that captures qualities, opinions, or behaviors (e.g., interview transcripts, photos, videos)
- Subjective and context-dependent, reflecting human experiences
- Rich in detail but harder to quantify compared to numerical data
- Sources: Interviews, focus groups, observations, open-ended survey responses, diaries, visual materials

Comparison with Quantitative

- Quantitative answers "how much?" or "how many?" (e.g., 70% passed the exam)
- Qualitative answers "why?" or "how?" (e.g., "Students felt the exam was fair")
- Complementary: Often used together for mixed-methods research

Analyzing Textual Data

- Learn systematic approaches to analyze text-based qualitative data
- Collect data by gathering transcripts, notes, or documents, ensuring ethical considerations (e.g., consent, anonymity)
- Read and re-read data to understand content, immersing in the context
- Assign labels to segments of text to identify themes or patterns (e.g., "stress" for "I feel overwhelmed by deadlines")
- Group codes into broader themes (e.g., "emotional responses" including stress, anxiety, satisfaction)
- Explain findings in context, linking to research questions or theoretical frameworks

Techniques: Analyzing Textual Data

- Content Analysis: Quantify frequency of themes (e.g., count how often "cost" is mentioned in customer reviews)
- Thematic Analysis: Identify recurring patterns or themes (e.g., fear of failure in student interviews)
- Narrative Analysis: Focus on stories or sequences, analyzing the structure and meaning (e.g., a patient's journey through treatment)

Analyzing Non-Textual Data

- Explore methods to analyze qualitative data beyond text, such as images or videos
- Types:
  - Images: Photos, drawings, advertisements (e.g., marketing visuals, protest photos)
  - Audio/Video: Recordings of interviews, focus groups, or events, capturing tone, gestures, expressions
  - Objects: Artifacts or physical items (e.g., historical tools, cultural artifacts)

Analysis Process

1. Description: Detail what is observed (e.g., in a photo, note bright colors, smiling faces, urban setting)
2. Contextualization: Link to broader context (e.g., cultural significance, historical period, social norms)
3. Interpretation: Infer meaning (e.g., a smile in a photo suggests happiness, but consider cultural variations)

Techniques

1. Visual Analysis: Examine composition, symbols, colors (e.g., red may symbolize danger or passion, depending on context).
2. Semiotics: Study signs and meanings, analyzing how visuals communicate (e.g., a logo's design conveys brand identity).
3. Discourse Analysis: Analyze spoken or visual communication patterns, considering power dynamics or narratives

Chapter 6

Ethical Issues in Research

Ashim Khadka

Research Techniques

1. Participants
2. The law and Research
3. Rights of People Directly Involved
   - Rights of participants
4. Responsibilities of an Ethical Researcher
5. Plagiarism and Intellectual Property
6. Environmental Responsibility
7. Code of Ethics
   - ACM Code of Ethics
   - IEEE Code of Ethics

Participants

- people directly or indirectly involved in your research
  - the people you interview, observe or ask to complete a questionnaire or supply you with documents
- should treat participants involved (directly or indirectly) in research: fairly and with honesty
  - ethical researcher
- require that you consider the ethical aspects of your research project when you design it
  - obtain ethical clearance before you start
- people affected by the research in any way will not be harmed and will be treated fairly and with dignity
- research $\rightarrow$ ethical $\rightarrow$ approved $\rightarrow$ start
- research $\rightarrow$ unethical (any part) $\rightarrow$ researcher will be asked to think again $\rightarrow$ redesign the project.

The law and Research

- research and behaviour throughout the project, from literature review to the final outcomes: must be legal
- find out about the law in your country concerning such things as:
  1. Data Protection Rights and Duties
  2. Prize Draws and Research Participation
  3. Intellectual Property Rights (IPR)
  4. Technology Restrictions
  5. Software Developers' Legal Liability

Data Protection Rights and Duties

- Individual Rights:
  - Individuals have the right to control their personal data
  - Researchers must handle data responsibly, ensuring privacy and security
  - Example: Suppose you're conducting a survey
    - You must inform participants how their data will be used and obtain their consent
- Duties of Organizations and Researchers:
  - Organizations must comply with data protection laws (such as GDPR in the EU)
  - Researchers should obtain informed consent, protect data, and handle it only for legitimate purposes
  - Example: If you're collecting health data for medical research, ensure strict confidentiality and compliance with relevant regulations

Prize Draws and Research Participation

- Offering prize draws can encourage participation, but ethical considerations apply:
  - Clearly state the terms and conditions
  - Ensure that participation is voluntary and not force or threat
  - Avoid biasing results by attracting participants solely for prizes
  - Example: In a study on consumer preferences, you offer a gift card as an incentive
    - communicates that participation is optional, and the prize doesn't affect the study outcome

Intellectual Property Rights (IPR)

- legal protections for creations such as inventions, artistic works, designs, symbols, and names
- for example, who owns the right to an image you want to use in your research, and who has the copyright of your own thesis or other publications or any software you produce

Technology Restrictions

- restrictions on the kinds of technology you are allowed to use and investigate
- Example
  - whether your country allows unrestricted access to the Internet
  - If developing a secure messaging app, verify that encryption complies with national regulations

Software Developers' Legal Liability

- the legal liability of software developers for the systems they design and create
- Example: If you develop a medical diagnosis app, rigorous testing and proper disclaimers are crucial to avoid harm to users.

Rights of People Directly Involved

- rights of the people most obviously affected by your research
  - the people you interview or observe, or ask to complete a questionnaire or supply you with documents
- the different terms used to refer to the people most directly impacted by your research
  1. **Research Subjects (Experiments)**
  2. **Research Respondents (Surveys)**
  3. **Research Informants or Members (Case Studies and Ethnography)**
  4. **Research Participants or Co-Researchers (Action Research)**
- all participants are humans
  - They have a **right** to be treated with dignity and, whenever possible, to gain some benefit from the research
  - You should consider carefully whether your work is both ethical and legal
    - ensure that your participants suffer no adverse consequences: physiological, psychological, social, political or economic

Research Subjects (Experiments)

- In experimental research, individuals who participate in controlled experiments are often referred to as research subjects.
- Example: If you're conducting a study on the effects of a new drug, the patients receiving the drug become your research subjects.

Research Respondents (Surveys)

- When collecting data through surveys or questionnaires, the individuals who respond are called research respondents.
- Example: If you distribute a survey about online shopping habits, the people who complete the survey are your research respondents.

Research Informants or Members (Case Studies and Ethnography)

- In case studies and ethnographic research, individuals who share information, experiences, or insights with the researcher are called informants or members
- Example: If you're studying a specific community's cultural practices, the community members who provide insights become your research informants.

Research Participants or Co-Researchers (Action Research)

- In action research, individuals actively engage in the research process. They may collaborate with the researcher, especially in more emancipatory forms of action research
- Example: If you're working with a group of teachers to improve classroom practices, they are both research participants and co-researchers.

Rights of participants

- Right not to participate
- Right to withdraw
- Right to give informed consent
- Right to anonymity
- Right to confidentiality

Right not to participate

- Participants have the right to decline participation in a study without any negative consequences
  - Researchers should respect their autonomy and not try to force them with threats
- Non-participation may affect the ability to complete research
  - that's researcher's problem, not theirs (participants)
- Example: If someone declines to be part of a survey on the shopping experience, their decision should be honoured.

Right to withdraw

- Participants can withdraw from a study at any time, even after initially agreeing to participate
- Researchers must inform participants of this right during the consent process
- Example: by declining to answer certain questions or by refusing to participate in some activities.

Right to give informed consent

- Participants have the right to receive clear and comprehensive information about the study before agreeing to participate
- Participants must be informed about:
  - the purpose of the research, why it is being undertaken and what benefits are expected from it
  - who is undertaking the research (name, address, contact details) and which organization is sponsoring it — either by funding it or overseeing and authorizing it
  - what will be involved (for example, interviews, completing a questionnaire and so on) and how long this is likely to take
  - whether they will receive any expenses, payment or incentive (for example, individual feedback on performance in a test, or a copy of the final research report);
  - how their data will be used and how the research findings will be disseminated.
- must also be informed that they have the right not to participate and the right to withdraw from the research at any time

Right to anonymity

- Participants in research have the right that their identity and location will be protected by disguise where necessary
- Researchers should use codes or pseudonyms to protect privacy
- Example: In an online survey, participants' names are not associated with their answers.

Right to confidentiality

- Researchers must keep participants' information confidential
- Data should be securely stored and accessible only to authorized personnel
- Researchers should not pass on sensitive information learned about participants to others in positions of authority (unless legally required)
- Example: If conducting interviews, avoid sharing specific quotes that could identify individual participants

- Is it ever justifiable to withhold from participants the aims of the research?
- Is it ever justifiable to carry out research when the subjects of the research cannot give informed consent?
- Is it justifiable to use people as the objects of research?
- Is it justifiable to carry out research that will solely or mainly benefit the researcher by, for example, leading to a PhD?
- Is it justifiable to carry out research when those funding the project retain the right to censor the findings?
- Is it justifiable to accept funding for health research from the tobacco industry?
- Is it justifiable to carry out research when the findings might be used to reduce staffing levels or withdraw treatment from certain categories of patients?

Responsibilities of an Ethical Researcher

- Ethical researchers can infer (conclude) their responsibilities from the rights of participants
  - researchers should respect participants' expectations of anonymity and confidentiality
  - they should not try to force (coerce) people into participating in the research
  - they should obtain informed consent and not deceive people about the research

Responsibilities of an Ethical Researcher

- No unnecessary intrusion
- Behave with integrity
- Follow appropriate professional codes of conduct
- No plagiarism
- Be an ethical reviewer

No unnecessary intrusion

- Researchers should respect the privacy and autonomy of individuals or groups they study
- Avoid unnecessary intrusion into their lives or personal spaces
- Obtain informed consent from participants before collecting data, especially in sensitive areas such as medical research or social studies
- Example, in designing questionnaires, researchers sometimes automatically insert questions asking about the participants' age, which respondents may find intrusive.

Behave with Integrity

- Integrity is crucial in research: involves honesty, transparency, and ethical behavior
- Researchers should accurately report their findings, even if those findings challenge their initial hypotheses
- ethical researcher also thinks about how the research findings will be used and tries to make sure they can do no harm

Follow Appropriate Professional Codes of Conduct

- Most professional bodies have produced codes of conduct
- Codes capture the profession's commitments and responsibilities, to help members make ethical decisions

No Plagiarism

- Plagiarism undermines the credibility of research
- Always attribute sources properly
  - When citing previous work, ideas, or data, follow the appropriate citation style (e.g., APA, MLA, etc.)

Be an Ethical Reviewer

- If you review research papers or grant proposals, evaluate them fairly and without bias
- Maintain confidentiality during peer review
- Provide constructive feedback to improve the quality of research

Data Privacy and Security

- Data security and privacy are essential in research data management to protect sensitive information from unauthorized access, breaches, or misuse
- Researchers need to use secure storage, and access controls to protect data throughout its lifecycle
  - Develop a data-collection plan for preserving participants' confidentiality
  - Acquire informed consent
  - Remove or modify personal identifiers in a dataset to prevent individuals from being identified
    - Key techniques include masking or generalizing data, suppression, and pseudonymization
- Compliance with privacy regulations such as GDPR and HIPAA is also essential when dealing with personal or health-related data

What are the best practices for ensuring data privacy in research?

Case Study

The Facebook-Cambridge Analytica scandal illustrates the misuse of personal data and highlights the need for stringent data privacy measures

- In 2018, it was revealed that Cambridge Analytica, a political consulting firm, had improperly accessed and used the personal data of approximately 87 million Facebook users without their explicit consent
  - Most users whose data was collected never gave consent or were even aware of it
- This data was harvested through a third-party app called "**This Is Your Digital Life**," which posed as a personality quiz
- While only about 270,000 people directly participated in the quiz and gave consent, the app exploited Facebook's API to access not just the participants' data but also the data of their friends and connections.

Case Study

- Cambridge Analytica used the data to create targeted political advertisements aimed at influencing elections, including the 2016 U.S.
- Presidential Election and the Brexit referendum in the UK. This raised concerns about manipulation of voter behavior and democracy.

Plagiarism

- Plagiarism occurs when someone uses another person's work, ideas, or expressions without giving proper credit or acknowledgment
  - Copying text, diagrams, or research data without citation
  - Presenting someone else's ideas as one's own
  - Failing to paraphrase adequately while using another person's research
- disrespects the original author's contributions and damages trust in the research community
- may breach copyright laws and academic codes of conduct

Types of Plagiarism

1. Direct Plagiarism: Word-for-word copying of someone else's work
2. Self-Plagiarism: Reusing your own previously published work without proper citation
3. Mosaic Plagiarism: Patching together phrases from multiple sources without proper acknowledgment
4. Accidental Plagiarism: Failing to cite a source due to oversight or lack of understanding

Intellectual Property

- Intellectual Property (IP) refers to creations of the mind, such as inventions, literary works, designs, symbols, names, or images
  - Copyright: Protects the original expression of ideas, such as articles or software
  - Patents: Grants exclusive rights to inventors for their inventions
  - Trademarks: Protects brand names and logos
  - Trade Secrets: Covers proprietary knowledge or practices that are kept confidential

Environmental Responsibility

- refers to conducting research activities in a way that minimizes negative environmental impacts and promotes sustainable practices
- involves making conscious decisions at every stage of the research process to reduce the consumption of resources, limit waste, and lower carbon footprints
  - Leveraging virtual meeting platforms (Zoom, Teams) reduces the environmental impact of flying researchers worldwide
- aligning with the United Nations Sustainable Development Goals (SDGs), particularly Goal 13 (Climate Action) and Goal 12 (Responsible Consumption and Production)
- High-performance computing (HPC) clusters used for simulations and machine learning consume vast energy
  - Leveraging green data centers powered by renewable energy
  - Scheduling tasks during low-energy-use periods
- Example: Google's Tensor Processing Units (TPUs) are designed to reduce energy use during large-scale AI computations, setting a benchmark for sustainable computational research

- Funding agencies increasingly require researchers to outline how their projects will minimize environmental impact
  - Example: The European Union's Horizon Europe program emphasizes environmental sustainability in its funding criteria
- Many universities have introduced "Green Labs" programs to promote sustainable research practices
  - Example: Harvard University's Green Labs Program provides resources for energy-efficient equipment and waste reduction strategies

Conflict of Interest (CoI)

- occurs when a researcher's personal, financial, or professional relationships interfere, or appear to interfere, with their ability to conduct research objectively
- undermine the integrity, credibility, and impartiality of the research process
- **Best Practices to Manage CoI**
  - Disclosure: Researchers must openly declare any potential conflicts of interest in publications, grant applications, or collaborations
  - Institutional Oversight: Universities and funding bodies should implement COI policies to review and address potential conflicts
  - Transparency: Maintain openness with collaborators, stakeholders, and the public.

Copyright

- form of intellectual property protection granted to creators of original works, such as books, software, music, and research articles
- gives the creator exclusive rights to reproduce, distribute, and display their work

Patent

- legal right granted to an inventor, giving them exclusive control over their invention for a specific period (usually 20 years)
- protects innovations that are novel, non-obvious, and useful

Code of Ethics

- set of principles that guide the behaviour of individuals or organizations, ensuring they act with integrity, fairness, and accountability
- enforced by internal policies or external regulatory bodies such as governments, professional associations, or industry watchdogs
- Example: General Data Protection Regulation (GDPR) in Europe, which will come into effect on May 25th, 2018
  - GDPR is created to enforce privacy rights and give people more control over their data
  - Penalties for violating it can be fines up to 20 million or 4% of annual global turnover - whichever is greater

Association for Computing Machinery (ACM) Code of Ethics

- provides guidelines for computing professionals to uphold integrity, fairness, and responsibility
- guides computing professionals in applying fundamental ethical principles to their conduct
  - Ethics in computing, responsible technology use, and social responsibility
- It is not a strict algorithm for solving ethical dilemmas but rather a foundation for ethical decision-making
- When addressing a specific issue, professionals must consider multiple principles, weighing their relevance to the situation
- consists of four key principles
  1. General Ethical Principles (Fundamental Ethical Responsibilities)
  2. Professional Responsibilities
  3. Professional Leadership Principles
  4. Compliance with the Code

General Ethical Principles (Fundamental Ethical Responsibilities)

- Contribute to society and human well-being
- Avoid harm to others
- Be honest and trustworthy
- Be fair and take action against discrimination
- Respect privacy and confidentiality

Professional Responsibilities

- Maintain high professional standards
- Know and respect laws and policies
- Honor contracts and agreements
- Ensure public understanding of computing systems

Professional Leadership Principles

- Create and maintain inclusive, respectful workplaces
- Support public policies that enhance social responsibility

Compliance with the Code

- Adhere to the code, encourage others, and take corrective actions when ethical violations occur

IEEE Code of Ethics

- Institute of Electrical and Electronics Engineers (IEEE) Code of Ethics promotes ethical decision-making for engineers and technologists
- focuses on Engineering ethics, professional conduct, and public safety
- includes the following principles
  1. Public Interest and Well-Being
  2. Honesty and Integrity
  3. Professional Responsibility
  4. Fairness and Non-Discrimination
  5. Environmental and Social Responsibility

Public Interest and Well-Being

- Prioritize the safety, health, and welfare of the public
- Disclose potential dangers in technology and engineering projects

Honesty and Integrity

- Be honest and transparent in claims, research, and project work
- Avoid deceptive practices

Professional Responsibility

- Maintain professional competence and improve technical knowledge
- Reject bribery and conflicts of interest
- Properly credit contributions and intellectual property

Fairness and Non-Discrimination

- Treat all individuals fairly, avoiding discrimination based on race, gender, religion, disability, etc.

Environmental and Social Responsibility

- Consider the environmental and social impact of engineering decisions.

Presentation of the Research

Chapter 7
Research Techniques

Ashim Khadka

Outline

1. Structure of Report Writing
   - Introduction
   - Background and Literature Review
   - Research Questions/Problem Statements
   - Methodology
   - Evaluations and Discussions
   - Conclusions
   - References
   - Abstract
2. Presentation

Structure of Report Writing

Structure of Report Writing

Introduction	What's Known? What's Unknown?
↓	
Background	What is the state of art in selected topic?
↓	
Methodologies	How do we show it?
↓	
Results	What are we showing now?
↓	
Discussion	What did we show?

Introduction

- **General introduction:** what is the research about
- **Summarize:** research questions/problems
- **Indicate:** why this is a worthwhile problem
- Give an overview of your contribution (solution, experiment and main results/conclusions)
- Can **general readers** easily understand the introduction?
- Did you **explain terms** to avoid readers' confusion?
- Did you **briefly introduce** your solutions and contributions?
- Did you **highlight** the novelty?

Background and Literature Review

- Give the background to let the readers know and understand your topic & the knowledge of this area
- The state of the art; related work, existing research
- Organize these by logical and reasonable ideas, not simply by time/author, etc
- Is the **background** clear enough?
- Is the **literature review** well organized and enough?

Research Questions/Problem Statements

- **Concise statement:** problems you want to address
- **Justified problem:** unanswered/not well solved based on the previous literature review
- **Thoroughly discuss:** why it is worthwhile to solve this problem?
- Will readers be convinced to **agree** with your view on the problems, about the originality, novelty and applicability?
- Did you refer to some **literature references** to convince your points?

Methodology

- **Main purpose:** to convince the examiners: you have answered the solved the problem by your own solutions
- It may **require several sections** to thoroughly explain the solutions, experiments and evaluations
- **Show** prior & current methodologies/work to your solution
- Can **readers** clearly understand your solutions?
- Did you **mention** the experimental conditions and limitations?

Evaluations and Discussions

- **Analyze:** your experimental results and evaluate your assumptions/performance of solutions
- **Focus** on the positive contributions of the results, also **explain** the negative results
- **Reasonably discuss** the limitations of your solution, **focus** on the scale of the applicability
- Did you **thoroughly**
  - evaluate the experimental results?
  - discuss the relevant issues, such as limitations, flaws, etc?

Conclusions

- **Conclusions:** short concise statements about the results of work you have done; must be directly relevant to the problem you raised in this research
  - How did you complete your promise?
- **Summary of contributions:** whether your results can contribute to the knowledge of your area; organize from most to least important
  - Original? Novel? Thorough? Meaningful?
- **Future research:** let other researchers pick up relevant work and follow your tracks
  - Feasible? Reasonable?

References

References

- Closely tied to the literature reviews in previous sections
- Examiners usually scan this list looking for important works in the field, which may be considered as a preliminary assessment
- Do not just list them, make sure you have referred them in the main body!
- **Organize** order of citation in your research: according to refereed reference style

Abstract

- General introduction of the background
- Briefly introduce the problem you want to address
- Summary of your solution and methodologies
- State a bird view of your results/conclusions
- Highlight your novel ideas and main contributions

Presentation

Presentation

- Introduction with problem statement
- Objectives
- Research question (if necessary)
- Summary of literature review with research gap select baseline literature
- Methodology: **Choice of Algorithms with reasons**
  - why do you choose and how does it differ from the state of the art?
- Block diagram with an explanation of each block
- Data collection (How many? and from where?)
- Expected outcome (with example)